

2015

ASIC LAB RECORD

BAVISKAR RADHESHYAM

14MVD1039



INDEX

SR.	Experiment Name	Date	Remark
1	Counter Design and Simulation Using nclaunch	06/01/2015	
2	Design of Adders	13/01/2015	
3	Design of simple Adding CPU	20/01/2015	
4	Netlist and SDC geraneration of Multiplier using RTL compiler	17/03/2015	
5	Low Design of Counter	24/03/2015	
6	Layout Design of Counter using Encounter tool	21/04/2015	
7	ASIC implementation of Johnson Counter	28/04/2015	

Date: 06.01.2015

ASIC Lab
Experiment No. 1
Design 4 Bit Counter

Radheshyam Bavisk
14MVD1039

AIM: To synthesize 4 bit counter RTL code and extract various reports for analysis of the circuit

TOOLS USED: Cadence RTL Compiler 64-bit

PROCEDURE:

1. Log into Unix Shell environment using **ssh** command via **terminal**
2. Source the required **sshrc** file from **/home/14mvd1039/cadence_DB** using **source sshrc** command
3. Navigate to the path **/home/14mvd1035/cadence_DB/Counter/rtl/bit4_counter** using command **cd /home/14mvd1035/cadence_DB/14MVD1035_HDL_Codes/bit4_counter**
4. Write the following timing constraint and save it as file **bit4_counter_constraint.g**:

```
create_clock -name clk_166mhz -period 6 -waveform {0 3} [get_ports "clk"]
set_clock_transition -rise 0.1 [get_clocks "clk_166mhz"]
set_clock_transition -fall 0.1 [get_clocks "clk_166mhz"]
set_clock_uncertainty 0.1 [get_ports "clk"]

set_input_delay -max 1.0 [get_ports "reset"] -clock [get_clocks "clk_166mhz"]
##set_input_delay -min 0.1 [get_ports "reset"] -clock [get_clocks "clk"]

set_output_delay -max 1.0 [get_ports "q"]
```
5. Invoke RTL Compiler tool using **rc -64** command inside the directory structure which holds the RTL code that needs to be synthesized (here **/home/14mvd1039/cadence_DBs/Counter/rtl/bit4_counter**)
6. The following commands are to be used for synthesis:
 - a. **set_attribute source_verbosity true**
 - b. **set_attribute information_level 9** – switch on all the **information** levels
 - c. **set_attribute library library/slow.lib** – load the required **library** file (slow.lib or fast.lib)
 - d. **read_hdl -v2001 bit4_counter.v** – read the RTL file for **syntax errors**
 - e. **set_top_design bit4_counter** – set the module in the RTL file as **top module**
 - f. **elaborate** – this creates a technology-independent schematic for bit4_counter
 - g. **read_sdc bit4_counter_constraint.g** – read the **clock constraint** file
 - h. **synthesize -to_mapped -effort low** – **synthesize** the RTL code with the effort setting as **low** (can either be **low** or **medium** or **high**)
 - i. **report_power >bit4_counter_power.rep** – save the power report to the file **bit4_counter_power.rep**
 - j. **report_area >bit4_counter_area.rep** – save the area report to the file **bit4_counter_area.rep**
 - k. **report_timing >bit4_counter_timing.rep** – save the timing report to the file **bit4_counter_timing.rep**
 - l. **report_qor >bit4_counter_qor.rep** – save the qor report to the file **bit4_counter_qor.rep**

- m. gui_show – show the RTL schematic of the circuit using **Encounter** tool
7. From the timing report, make sure that timing slack is not negative. If it is, modify the timing constraint file (clock signal), follow the entire set of steps mentioned above and note down the timing slack again. Follow this procedure, till the timing slack obtained is positive

VERILOG CODE:

4BIT COUNTER---

```
module counter(clk,rst,q);
input clk,rst;
output reg [3:0] q;
always @(posedge clk)
begin
if(rst==1'b1)
q<=4'b0;
else
q<=q+1'b1;
end
endmodule
```

4BIT COUNTER TEST BENCH---

```
module counter_tb();
reg clk=1'b0;
reg rst;
wire [3:0] q;
counter z1(clk,rst,q);
always #5 clk=~clk;
initial
begin
rst=1'b1;
#20 rst=1'b0
```

```

end
initial
begin
$monitor("clk=%b,rst=%b,q=%b",clk,rst,q);
#400
$finish;
end
endmodule

```

RTL SCHEMATIC:

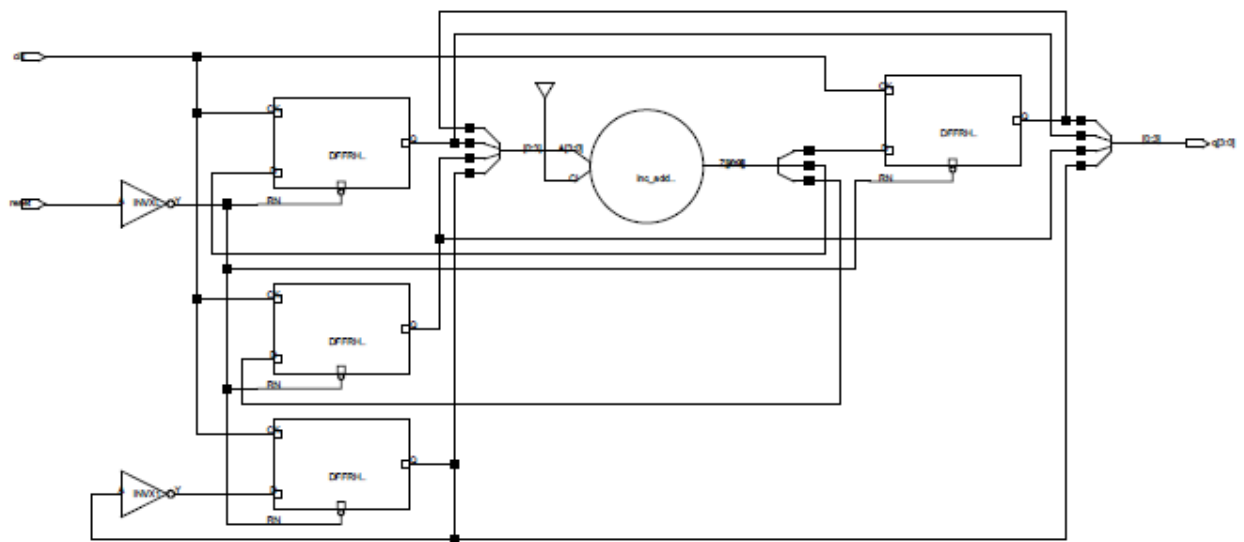


Fig 1(a).

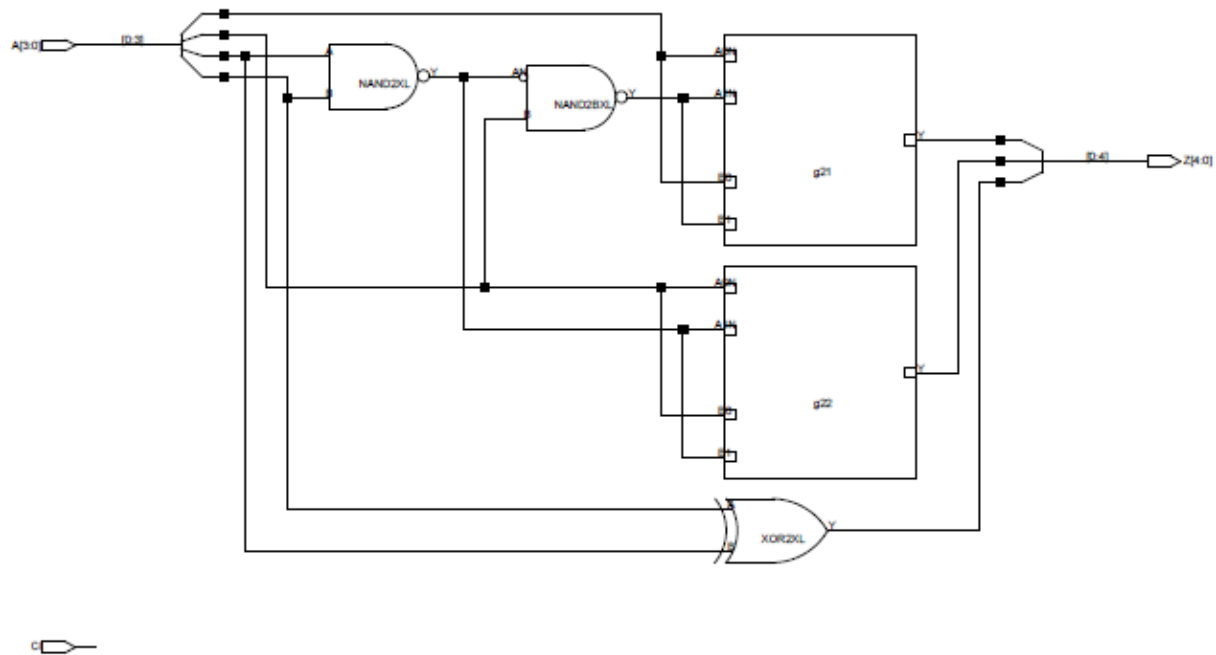


Fig 1(b).

REPORTS:

Area

```

=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Jan 27 2015  03:46:50 PM
Module:           bit4_counter
Technology library: tsmc18 1.0
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====
  
```

Instance	Cells	Cell Area	Net Area	Wireload
bit4_counter	11	389	0	<none> (D)
inc_add_10_10_1	5	96	0	<none> (D)

(D) = wireload is default in technology library

Power

```
=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Jan 27 2015  03:43:45 PM
Module:           bit4_counter
Technology library: tsmc18 1.0
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====
```

		Leakage	Dynamic	Total
Instance	Cells	Power(nW)	Power(nW)	Power(nW)
bit4_counter	11	12.810	53378.346	53391.156
inc_add_10_10_1	5	3.699	2504.498	2508.197

Timing

```
=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Jan 27 2015  04:24:15 PM
Module:           bit4_counter
Technology library: tsmc18 1.0
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====
```

Pin	Type	Fanout	Load (fF)	Slew (ps)	Delay (ps)	Arrival (ps)
(clock clk_166mhz)	launch					0 R
q_reg[0]/CK		100				0 R
q_reg[0]/Q	DFFRHQX1	4	10.6	285	+452	452 R
g200/B		+0		452		
g200/Y	NAND2X1	2	8.1	127	+103	554 F
q_reg[2]/SE	SDDFRHQX1				+0	554
q_reg[2]/CK	setup	100		+747		1302 R
(clock clk_166mhz)	capture					6000 R
uncertainty		-100				5900 R

Cost Group : 'clk_166mhz' (path_group 'clk_166mhz')
Timing slack : 4598ps
Start-point : q_reg[0]/CK
End-point : q_reg[2]/SE

QOR

```
=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Jan 27 2015 04:23:55 PM
Module:           bit4_counter
Technology library: tsmc18 1.0
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====
```

Timing

```
-----
Cost   Critical   Violating
Group  Path Slack TNS   Paths
-----
default  No paths  0
clk_166mhz 4598.5 0      0
clk      No paths  0
-----
Total           0      0
```

Instance Count

```
-----
Leaf Instance Count      10
Sequential Instance Count 4
Combinational Instance Count 6
Hierarchical Instance Count 0
```

Area & Power

```
-----
Total Area                369.230
Cell Area                 369.230
Leakage Power             9.999 nW
Dynamic Power             55250.703 nW
Total Power               55260.703 nW
```

Max Fanout	4 (clk)
Min Fanout	1 (n_0)
Average Fanout	2.4
Terms to net ratio	3.0
Terms to instance ratio	3.6
Runtime	6 seconds
Hostname	localhost.localdomain

Date: 13.01.2015

ASIC Lab
Experiment No. 2

**Compare the RCA(ripple carry adder) and
CLA(Carry Look Ahead) adder**

**Radheshyam Baviskar
14MVD1039**

AIM: To synthesize and compare the two adders ripple carry adder and Carry look ahead adder .

TOOLS USED: Cadence RTL Compiler 64-bit

PROCEDURE:

1. Log into Unix Shell environment using **ssh** command via **terminal**
2. Source the required **sshrc** file from **/home/14mvd1039/cadence_DB** using **source sshrc** command
3. Navigate to the path **/home/14mvd1035/cadence_DB/Counter/rtl** using command **cd /home/14mvd1035/cadence_DB/Counter/rtl**
4. Invoke RTL Compiler tool using **rc -64** command inside the directory structure which holds the RTL code that needs to be synthesized (here **/home/14mvd1035/cadence_DBs/Counter/rtl/**)
5. The following commands are to be used for synthesis:
 - a. **set_attribute source_verbose true**
 - b. **set_attribute information_level 9** – switch on all the **information** levels
 - c. **set_attribute library library/slow.lib** – load the required **library** file (slow.lib or fast.lib)
 - d. **read_hdl -v2001 bit4_counter.v** – read the RTL file for **syntax errors**
 - e. **set_top_design bit4_counter** – set the module in the RTL file as **top module**
 - f. **elaborate** – this creates a technology-independent schematic for bit4_counter
 - g. **read_sdc bit4_counter_constraint.g** – read the **clock constraint** file
 - h. **synthesize -to_mapped -effort low** – **synthesize** the RTL code with the effort setting as **low** (can either be **low** or **medium** or **high**)
 - i. **report power >**
 - j. **report area >**
 - k. **report timing >**
 - l. **report qor >**
 - m. **gui_show** – show the RTL schematic of the circuit using **Encounter** tool
6. From the timing report, make sure that timing slack is not negative. If it is, modify the timing constraint file (clock signal), follow the entire set of steps mentioned above and note down the timing slack again. Follow this procedure, till the timing slack obtained is positive

VERILOG CODE:

RIPPLE CARRY ADDER USING FULL ADDER INSTANTIATION ---

FULL ADDER-----

```
module fa(a,b,cin,sum,carry);  
input a,b,cin;  
output sum,carry;  
assign sum= a^b^cin;  
assign carry= (a &b)|(a&cin)|(b&cin);  
endmodule
```

MAIN RCA-----

```
module rca(a,b,cin,sum,carry);  
input [3:0]a,b;  
input cin;  
output [3:0]sum;  
output carry;  
wire [2:0]c;  
fa t1(a[0],b[0],cin,sum[0],c[0]);  
fa t2(a[1],b[1],c[0],sum[1],c[1]);  
fa t3(a[2],b[2],c[1],sum[2],c[2]);  
fa t4(a[3],b[3],c[2],sum[3],carry);  
endmodule
```

RCA TEST BENCH---

```
module rca_test;  
reg [3:0]a,b;  
reg cin;  
wire [3:0]sum;  
wire carry;  
rca dut(a,b,cin,sum,carry);
```

```

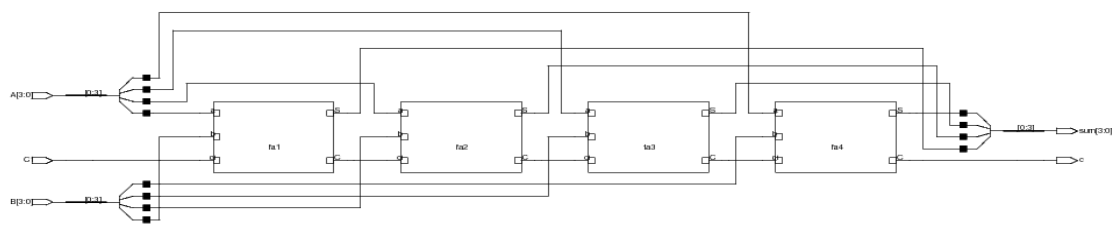
initial
begin
a=4'b1010;
b=4'b1001;
c0= 1'd0;
#100
a=4'b1110;
b=4'b1011;
c0= 1'd0;
#100
a=4'b1111;
b=4'b1111;
c0= 1'd0;
end

//$monitor("a=%b,b=%b,c0=%b,s=%b,c=%b",a,b,c0,s,c);
//#400 $finish;

endmodule

```

RTL SCHEMATIC(RCA):



REPORTS:

rc:/> **report gates**

```
=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Feb 09 2015 05:14:48 PM
Module:           ripple_adder
Technology library: tsmc18 1.0
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====
```

Gate	Instances	Area	Library
INVXL	4	26.611	tsmc18
NAND2XL	4	39.917	tsmc18
NOR2XL	4	39.917	tsmc18
OAI21XL	8	106.445	tsmc18

total	20	212.890	

Type	Instances	Area	Area %
inverter	4	26.611	12.5
logic	16	186.278	87.5

total	20	212.890	100.0

rc:/> **report timing**

Warning : Possible timing problems have been detected in this design. [TIM-11]
: The design is 'ripple_adder'.
: Use 'report timing -lint' for more information.

```
=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Feb 09 2015 05:14:55 PM
Module:           ripple_adder
Technology library: tsmc18 1.0
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====
```

Pin	Type	Fanout	Load	Slew	Delay	Arrival
-----	------	--------	------	------	-------	---------

			(fF)	(ps)	(ps)	(ps)	
a[0]	in port	2	5.5	0	+0	0	R
fa1/a							
g75/B				+0	0		
g75/Y	NOR2XL	2	6.8	99	+68	68	F
g73/A0				+0	68		
g73/Y	OAI21XL	2	6.0	304	+214	282	R
fa1/c							
fa2/ci							
g76/A				+0	282		
g76/Y	INVXL	1	3.4	89	+70	352	F
g73/A0				+0	352		
g73/Y	OAI21XL	2	6.0	304	+212	564	R
fa2/c							
fa3/ci							
g76/A				+0	564		
g76/Y	INVXL	1	3.4	89	+70	633	F
g73/A0				+0	633		
g73/Y	OAI21XL	2	6.0	304	+212	845	R
fa3/c							
fa4/ci							
g76/A				+0	845		
g76/Y	INVXL	1	3.4	89	+70	915	F
g73/A0				+0	915		
g73/Y	OAI21XL	1	0.0	149	+123	1038	R
fa4/c							
c	out port			+0	1038		R

Timing slack : UNCONSTRAINED

Start-point : a[0]

End-point : c

rc:/> **report area**

```

=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Feb 09 2015 05:15:00 PM
Module:           ripple_adder
Technology library: tsmc18 1.0
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====

```


Instance	Cells	Cell Area	Net Area	Wireload
ripple_adder	20	213	0	<none> (D)
fa4	5	53	0	<none> (D)
fa3	5	53	0	<none> (D)
fa2	5	53	0	<none> (D)
fa1	5	53	0	<none> (D)

(D) = wireload is default in technology library

rc:/> **report power**

```

=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Feb 09 2015 05:15:11 PM
Module:           ripple_adder
Technology library: tsmc18 1.0
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====

```

		Leakage	Dynamic	Total
Instance	Cells	Power(nW)	Power(nW)	Power(nW)
ripple_adder	20	3.714	17803.381	17807.096
fa1	5	0.929	1475.788	1476.716
fa2	5	0.929	1691.849	1692.777
fa3	5	0.929	1828.419	1829.347
fa4	5	0.929	1416.394	1417.323

VERILOG CODE:

CARRY LOOK AHEAD ADDER -----

```
module cla(a,b,cin,s,cout);
input[3:0] a,b;
input cin;
output cout;
output[3:0] s;
wire[3:0] g,p;
wire[13:0] z;
xor ((p[0]),(a[0]),(b[0]));
and ((g[0]),(a[0]),(b[0]));
xor ((p[1]),(a[1]),(b[1]));
and ((g[1]),(a[1]),(b[1]));
xor ((p[2]),(a[2]),(b[2]));
and ((g[2]),(a[2]),(b[2]));
xor ((p[3]),(a[3]),(b[3]));
and ((g[3]),(a[3]),(b[3]));
xor ((s[0]),(cin),(p[0]));
and ((z[0]),(cin),(p[0]));
or ((z[1]),(z[0]),(g[0]));
xor ((s[1]),(z[1]),(p[1]));
and ((z[2]),(cin),(p[0]),(p[1]));
and ((z[3]),(g[0]),(p[1]));
or ((z[4]),(z[2]),(z[3]),(g[1]));
xor ((s[2]),(z[4]),(p[2]));
and ((z[5]),(cin),(p[0]),(p[1]),(p[2]));
and ((z[6]),(g[0]),(p[1]),(p[2]));
and ((z[7]),(g[1]),(p[2]));
or ((z[8]),(z[5]),(z[6]),(z[7]),(g[2]));
xor ((s[3]),(z[8]),(p[3]));
and ((z[9]),(cin),(p[0]),(p[1]),(p[2]));
and ((z[10]),(g[0]),(p[1]),(p[2]));
and ((z[11]),(g[1]),(p[2]));
or ((z[12]),(z[9]),(z[10]),(z[11]),(g[2]));
and ((z[13]),(z[12]),(p[3]));
or ((cout),(z[13]),(g[3]));
endmodule
```

TESTBENCH:

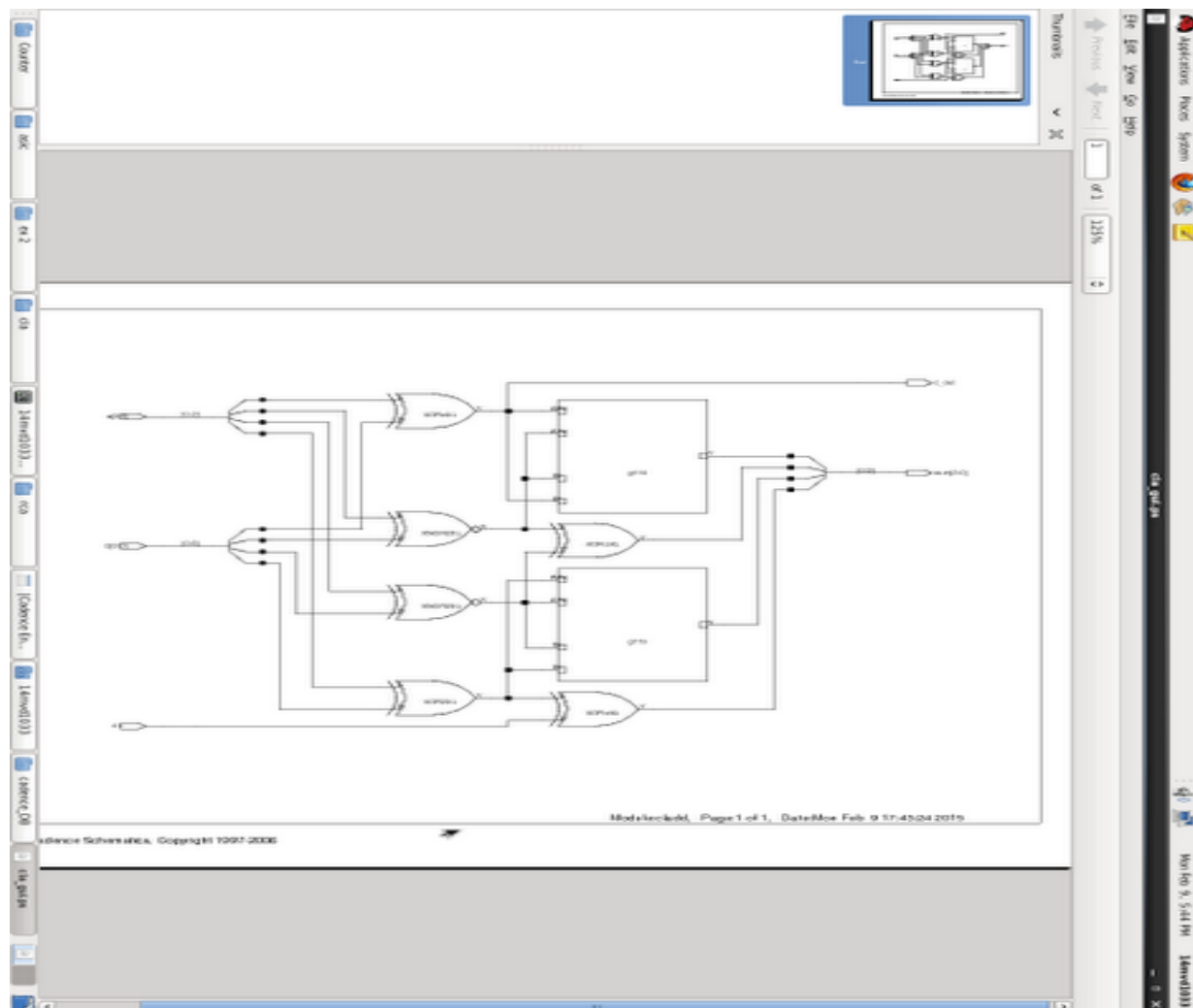
```
module cla_test;
reg [3:0] a,b;
reg cin;
wire [3:0] s;
wire cout;
```

```

cla t(a,b,cin,s,cout);
initial
begin
a=4'b1010;
b=4'b1001;
cin=0;
#50
a=4'b1110;
b=4'b1011;
#50
a=4'b1111;
b=4'b1111;
end
endmodule

```

RTL SCHEMATIC(CLA):-



INFERENCE :-

TABLE OF COMPARISION

FACTORS/ADDER	RCA	CLA
POWER(nW)	17807.096(nW)	14500.74(nW)
CELLS	50	63
AREA	246	296

From comparing with respect to designs and above table results, though the complexity is less in the RCA ,CLA has the advantage of good power, less area and lesser delay.

Date-20/02/2015

ASIC Lab
Experiment No. 3

ADDING CPU

Radheshyam Baviskar
14MVD1039

1. Write HDL codes for individual modules, AC,IR,PC,ALU

2. Write the datapath and controlpath based on architecture design
3. Write the Top-level design module
4. Write the testbench to test the top level module
5. Synthesize and get the reports.

HDL codes:

AC

```
module AC ( input [7:0] data_in, input load, clk,
outputreg [7:0] data_out );
always @( posedgeclk )
if( load ) data_out<= data_in;
endmodule
```

IR

```
module IR ( input [7:0] data_in, input load, clk,
outputreg [7:0] data_out );
always @( posedgeclk )
if ( load ) data_out<= data_in;
endmodule
```

PC

```
module PC ( input [5:0] data_in, input load, inc, clr, clk,
outputreg [5:0] data_out );
always @( posedgeclk )
if(clr ) data_out<= 6'b000_000;
else if( load ) data_out<= data_in;
else if( inc ) data_out<= data_out + 1;
endmodule
```

ALU

```
module ALU ( input [7:0] a, b, input pass, add,
outputreg[7:0] alu_out );
always @(a or b or pass or add)
if (pass) alu_out = a;
else if (add) alu_out = a + b;
else alu_out = 0;
endmodule
```

DATAPATH

```
moduledatapath ( input ir_on_adr, pc_on_adr, dbus_on_data,
```

```

data_on_dbus, ld_ir, ld_ac, ld_pc,
inc_pc, clr_pc, pass, add, alu_on_dbus,
clk,
output [5:0] adr_bus,
output [1:0] op_code,
inout [7:0] data_bus );
wire [7:0] dbus, ir_out, a_side, alu_out;
wire [5:0] pc_out;
IR ir( .data_in(dbus), .load(ld_ir), .clk(clk), .data_out(ir_out) );
PC pc( .data_in(ir_out[5:0]), .load(ld_pc), .inc(inc_pc),
.clr(clr_pc), .clk(clk), .data_out(pc_out) );
AC ac( .data_in(dbus), .load(ld_ac), .clk(clk), .data_out(a_side) );
ALU alu( .a(a_side), .b({2'b00,ir_out[5:0]}), .pass(pass), .add(add),
.alu_out(alu_out) );

assignadr_bus = ir_on_adr ? ir_out[5:0] : (pc_on_adr ? pc_out :
6'bzz_zzzz);
//assign adr_bus = pc_on_adr ?pc_out : 6'bzz_zzzz;
assigndbus = alu_on_dbus ? alu_out : (data_on_dbus ? data_bus :
8'bzzzz_zzzz);
assigndata_bus = dbus_on_data ? dbus : 8'bzzzz_zzzz;
//assign dbus = data_on_dbus ?data_bus : 8'bzzzz_zzzz;

assignop_code = ir_out[7:6];
endmodule

```

CONTROLPATH

```

modulecontrolpath ( input reset, clk, input [1:0] op_code,
outputregrd_mem, wr_mem, ir_on_adr,
pc_on_adr, dbus_on_data,
data_on_dbus, ld_ir, ld_ac,
ld_pc, inc_pc, clr_pc,
pass, add, alu_on_dbus);
reg [1:0] present_state, next_state;

parameter Reset= 2'b00;
parameter Fetch = 2'b01;
parameter Decode = 2'b10;
parameter Execute = 2'b11;
always @( posedgeclk )
if( reset ) present_state<= Reset;
elsepresent_state<= next_state;
always @( present_state or reset ) begin : Combinational
rd_mem=1'b0; wr_mem=1'b0; ir_on_adr=1'b0; pc_on_adr=1'b0;
dbus_on_data=1'b0; data_on_dbus=1'b0; ld_ir=1'b0;
ld_ac=1'b0; ld_pc=1'b0; inc_pc=1'b0; clr_pc=1'b0;
pass=0; add=0; alu_on_dbus=1'b0;

```



```

case ( present_state )
Reset : begin next_state = reset ? Reset : Fetch;
clr_pc = 1;
end // End `Reset

Fetch : begin next_state = Decode;
pc_on_adr = 1; rd_mem = 1; data_on_dbus = 1;
ld_ir = 1; inc_pc = 1;
end // End `Fetch
Decode :next_state = Execute; // End `Decode
Execute: begin next_state = Fetch;
case(op_code )
2'b00: begin
ir_on_adr = 1; rd_mem = 1;
data_on_dbus = 1; ld_ac = 1;
end
2'b01: begin
pass = 1;
ir_on_adr = 1; alu_on_dbus = 1;
dbus_on_data = 1; wr_mem = 1;
end

2'b10: ld_pc = 1;
2'b11: begin
add = 1; alu_on_dbus = 1; ld_ac = 1;
end
endcase
end // End `Execute
default :next_state = Reset;
endcase
end
endmodule

```

TOPMODULE

```

moduleAddingCPU (input reset, clk,
output [5:0] adr_bus, output rd_mem, wr_mem,
inout [7:0] data_bus);
wireir_on_adr, pc_on_adr, dbus_on_data, data_on_dbus, ld_ir, ld_ac,
ld_pc, inc_pc, clr_pc, pass, add, alu_on_dbus;
wire [1:0] op_code;

controlpath cu ( .reset(reset), .clk(clk), .op_code(op_code),
.rd_mem(rd_mem), .wr_mem(wr_mem), .ir_on_adr(ir_on_adr),
.pc_on_adr(pc_on_adr), .dbus_on_data(dbus_on_data),
.data_on_dbus(data_on_dbus), .ld_ir(ld_ir), .ld_ac(ld_ac),
.ld_pc(ld_pc), .inc_pc(inc_pc), .clr_pc(clr_pc), .pass(pass),
.add(add), .alu_on_dbus(alu_on_dbus));

```

```

datapathdp ( .ir_on_adr(ir_on_adr), .pc_on_adr(pc_on_adr),
.dbus_on_data(dbus_on_data), .data_on_dbus(data_on_dbus),
.ld_ir(ld_ir), .ld_ac(ld_ac), .ld_pc(ld_pc), .inc_pc(inc_pc),
.clr_pc(clr_pc), .pass(pass), .add(add), .alu_on_dbus(alu_on_dbus),
.clk(clk), .adr_bus(adr_bus), .op_code(op_code), .data_bus(data_bus)
);

```

```

endmodule

```

TESTBENCH(using task convert):

```

moduleTest_AddingCPU;
reg reset=1, clk=0;
wire [5:0] adr_bus;
wirerd_mem, wr_mem;
wire [7:0] data_bus;
reg [7:0] mem_data=0;
reg control=0;
integerHexFile, check;

```

```

AddingCPUAddingCPU_UUT (.adr_bus(adr_bus), .rd_mem(rd_mem),
.wr_mem(wr_mem), .data_bus(data_bus), .reset(reset),.clk(clk));

```

```

initial begin
    forever #10 clk = ~clk;
end

```

```

initial begin
    Convert;
    HexFile = $fopen ("HexadecimalFile.mem", "r+");
    #25 reset=1'b0;
    #405 $fclose (HexFile);
    $stop;

    forever @ (posedgeclk)
    begin
        control = 0;
        #1;
        if (rd_mem)
        begin
            #1;
            check = $fseek (HexFile, 4 * adr_bus, 0);
            check = $fscanf (HexFile, "%h", mem_data);
            control = 1;
        end

        if (wr_mem)
        begin
            #1;
            check = $fseek (HexFile, 4 * adr_bus, 0);

```

```

        $fwrite (HexFile, "%h", data_bus);
        $fflush (HexFile);
    end
end
end

assigndata_bus = (control) ? mem_data: 8'hZZ;

task Convert;
begin: block

    reg [5: 0] addr;
    reg [3 * 8: 1] opCode;
    reg [7: 0] data, writeData;
    regJustData;
    integeri, HexFile, InstFile, check;

    HexFile = $fopen ("HexadecimalFile.mem");

    for (i = 0; i < 64; i = i + 1) $fwrite (HexFile, "00\n");

    $fflush (HexFile); $fclose (HexFile);
    InstFile = $fopen ("InstructionFile.mem", "r");
    HexFile = $fopen ("HexadecimalFile.mem", "r+");

    while ($fscanf (InstFile, "%h", addr) != -1)
    begin
        check = $fseek (HexFile, addr * 4, 0);
        check = $fgets (opCode, InstFile);
        JustData = 0;

        case (opCode)
            "lda": writeData[7: 6] = 0;
            "sta": writeData[7: 6] = 1;
            "jmp": writeData[7: 6] = 2;
            "add": writeData[7: 6] = 3;
            "::::": begin
                JustData = 1;
                check = $fscanf (InstFile, "%h", writeData);
            end
            default: begin
                JustData = 1;
                check = $fscanf (InstFile, "%h", writeData);
            end
        endcase

        if(JustData == 0)
        begin
            check = $fscanf (InstFile, "%h", data);
            writeData[5: 0] = data[5: 0];
        end
    end
end

```

```

        $fwrite(HexFile, "%h", writeData);
    end

    $fflush (HexFile); $fclose (HexFile); $fclose(InstFile);
end
endtask

endmodule

```

TESTBENCH(without using task convert):

```

module adding_cpu_testbench;

    wire [5:0] adr_bus;
    wire rd_mem;
    wire wr_mem;
    wire [7:0] data_bus;
    reg reset;
    reg clk;

    reg [7:0] data_bus_temp;
    reg oe;

    AddingCPUAddingCPU_uut (.adr_bus(adr_bus), .rd_mem(rd_mem),
        .wr_mem(wr_mem), .data_bus(data_bus), .reset(reset), .clk(clk));

    assign data_bus = oe ? 8'bZ : data_bus_temp;

    initial begin
        clk = 1;
        forever #10 clk = ~clk;
    end

    initial begin
        reset = 1;
        #40 reset = 0; oe = 0;

        #20 data_bus_temp = 8'b00111111; //lda
        #60 data_bus_temp = 8'b11110000; //add
        #60 data_bus_temp = 8'b01111111; //sta
        #40 oe = 1;
        #20 oe = 0; data_bus_temp = 8'b10111111; //jump

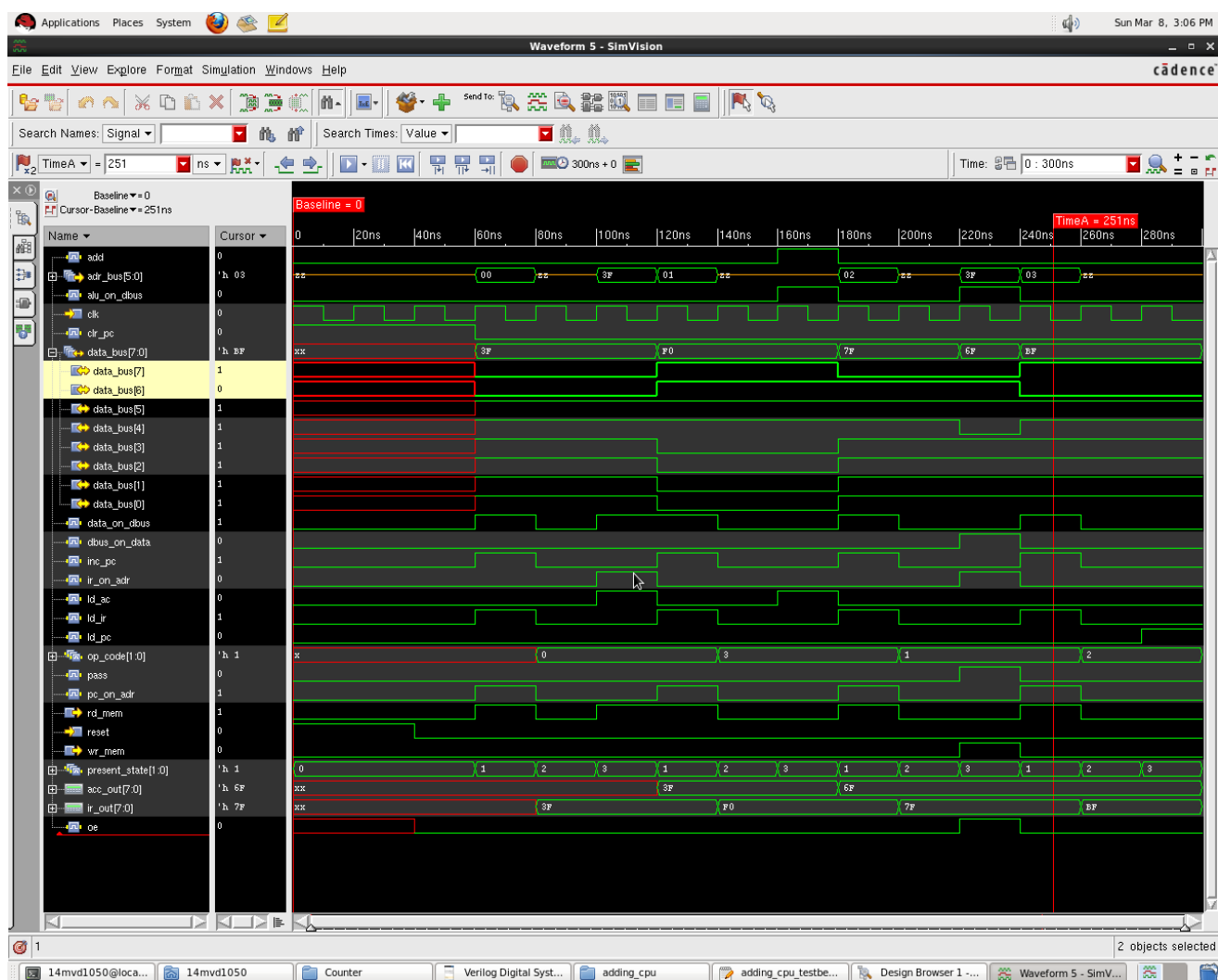
        #60 $stop;
        #60 $finish;
    end

end

endmodule

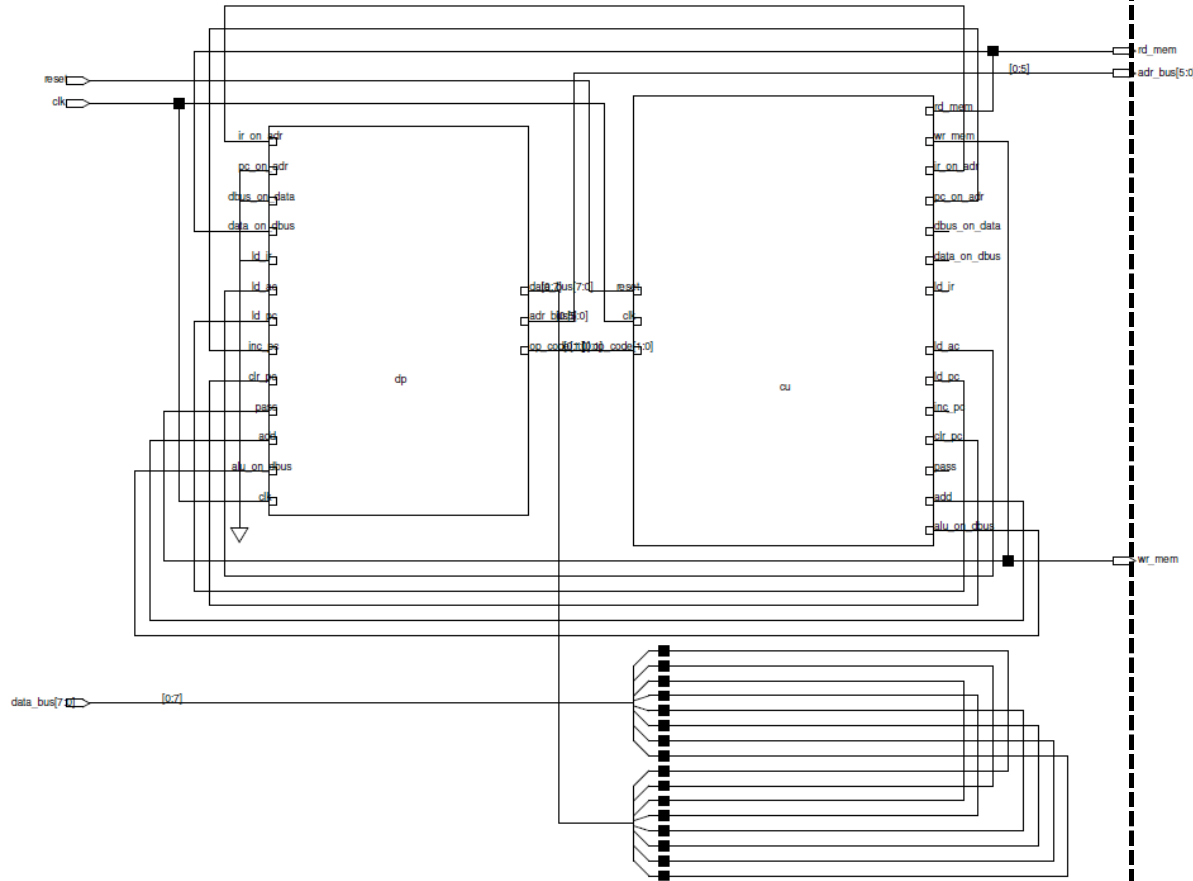
```

SIMULATION RESULT:

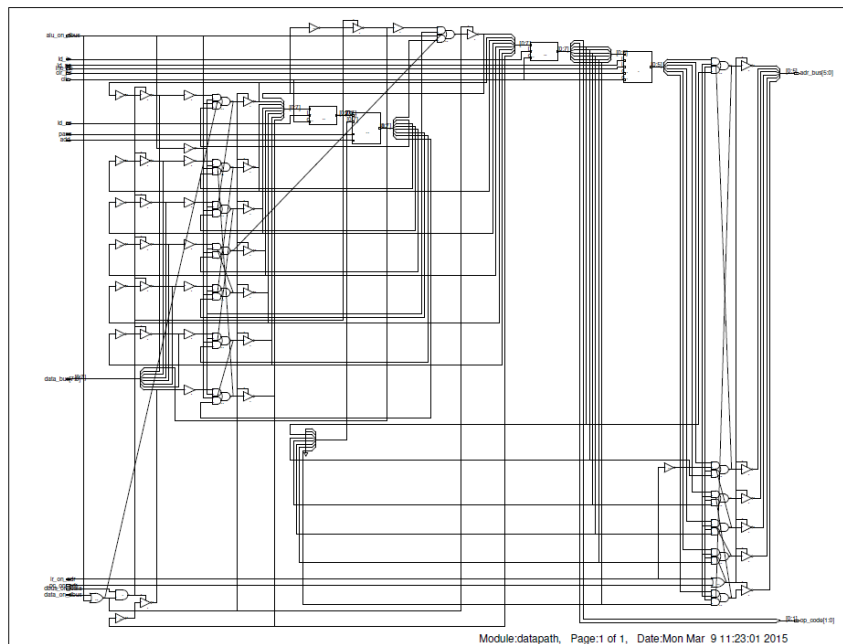


Fig(A

Fig(A

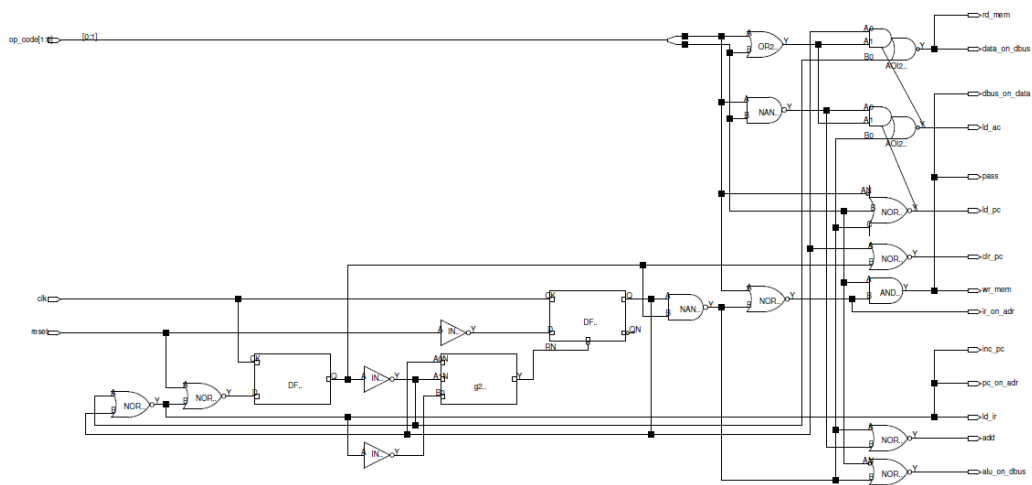


Fig(a)Top level



Cadence Schematics, Copyright 1997-2006

Fig(b) Datapath



Fig©controlpath

Reports:

1. Power

```
=====
Generated by:          Encounter(R) RTL Compiler v09.10-p104_1
Generated on:          Mar 10 2015  10:46:01 AM
Module:                AddingCPU
Technology library:    tsmc18 1.0
Operating conditions:  slow (balanced_tree)
Wireload mode:         enclosed
Area mode:             timing library
=====
```

Instance	Cells	Leakage Power (nW)	Dynamic Power (nW)	Total Power (nW)
AddingCPU	152	112.878	274465.522	274578.400
dp	132	105.614	236207.768	236313.382
alu	26	29.503	16796.988	16826.491
add_5_27	9	22.462	16796.988	16819.450
pc	33	24.229	52379.695	52403.923
inc_add_6_37_1	9	6.120	3839.515	3845.635
ac	8	17.251	69934.157	69951.409
ir	8	17.251	76247.836	76265.087
cu	20	7.264	20243.872	20251.136

2. Gates

```
=====
Generated by:          Encounter(R) RTL Compiler v09.10-p104_1
Generated on:          Mar 10 2015  10:46:01 AM
Module:                AddingCPU
Technology library:    tsmc18 1.0
Operating conditions:  slow (balanced_tree)
Wireload mode:         enclosed
Area mode:             timing library
=====
```

Gate	Instances	Area	Library
ADDFX2	1	69.854	tsmc18
ADDHXL	1	36.590	tsmc18
AND2X2	1	13.306	tsmc18
AOI21X1	2	26.611	tsmc18
AOI21XL	1	13.306	tsmc18
AOI22X1	19	316.008	tsmc18
CLKBUFXL	8	106.445	tsmc18
CMPR32X1	4	279.418	tsmc18

DFFHQX1	8	425.779	tsmc18
INVX1	16	106.445	tsmc18
NAND2BX1	3	39.917	tsmc18
NAND2X1	6	59.875	tsmc18
NAND2XL	8	79.834	tsmc18
NOR2BX1	2	26.611	tsmc18
NOR2X1	5	49.896	tsmc18
NOR2XL	2	19.958	tsmc18
NOR3BX1	1	19.958	tsmc18
NOR3X1	3	39.917	tsmc18
OAI221XL	1	23.285	tsmc18
OAI2BB1X1	13	216.216	tsmc18
OAI2BB2X1	3	69.854	tsmc18
OAI2BB2XL	2	46.570	tsmc18
OR2X2	2	26.611	tsmc18
SDDFFHQX1	16	1064.448	tsmc18
TBUFIX1	22	292.723	tsmc18
XOR2X1	2	53.222	tsmc18

total	152	3522.658	

Type	Instances	Area	Area %

sequential	24	1490.227	42.3
inverter	16	106.445	3.0
buffer	8	106.445	3.0
tristate	22	292.723	8.3
logic	82	1526.818	43.3

total	152	3522.658	100.0

3. Area

```

=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Mar 10 2015  10:46:01 AM
Module:           AddingCPU
Technology library: tsmc18 1.0
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====

```

Instance	Cells	Cell Area	Net Area	Wireload

AddingCPU	152	3523	0	<none> (D)
dp	132	3230	0	<none> (D)
pc	33	755	0	<none> (D)

	inc_add_6_37_1	9	170	0	<none> (D)
alu		26	672	0	<none> (D)
	add_5_27	9	446	0	<none> (D)
ir		8	532	0	<none> (D)
ac		8	532	0	<none> (D)
cu		20	293	0	<none> (D)

(D) = wireload is default in technology library

4. Timing

```

=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Mar 10 2015  10:46:01 AM
Module:           AddingCPU
Technology library: tsmc18 1.0
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====

```

Pin	Type	Fanout	Load	Slew	Delay
Arrival		(fF)	(ps)	(ps)	
(ps)					

(clock clk_100MHz)	launch				
0 R					
dp					
ir					
data_out_reg[0]/CK		100			0
R					
data_out_reg[0]/Q	SDDFFHQX1	4	15.4	228	+422 422
R					
ir/data_out[0]					
alu/b[0]					
add_5_27/B[0]					
g142/B				+0	422
g142/CO	ADDHXL	1	6.9	170	+218 641 R
g141/A					+0
641					
g141/CO	CMR32X1	1	6.9	188	+511 1151 R
g140/A					+0
1151					
g140/CO	CMR32X1	1	6.9	188	+513 1664 R
g139/A					+0
1664					
g139/CO	CMR32X1	1	6.9	188	+513 2177 R
g138/A					+0
2177					

g138/CO	CMPR32X1	1	6.9	188	+513	2690 R
g137/A						+0
2690						
g137/CO	ADDFX2	2	9.2	140	+534	3223 R
g136/B					+0	3223
g136/Y	NAND2X1	2	6.6	112	+86	3309 F
g134/A1N					+0	3309
g134/Y	OAI2BB2X1	1	1.8	95	+197	3506 F
add_5_27/Z[7]						
g149/A0N					+0	3506
g149/Y	OAI2BB1X1	1	4.3	98	+218	3724
F						
alu/alu_out[7]						
g179/B1						+0
3724						
g179/Y	AOI22X1	1	4.5	263	+230	
3954 R						
g36/A						+0
3954						
g36/Y	TBUFIX1	3	11.5	136	+119	
4074 F						
g174/A						+0
4074						
g174/Y	INVX1	1	4.5	106	+90	
4163 R						
g43/A						+0
4163						
g43/Y	TBUFIX1	3	5.6	74	+74	
4237 F						
dp/data_bus[7]						
data_bus[7]	out port					+0
4237 F						
(AddingCPU_constraint_line_9)	ext delay					+1000
5237 F						
- - - - -						
- - -						
(clock clk_100MHz)	capture					
10000 R						

```

-----
Cost Group   : 'clk_100MHz' (path_group 'clk_100MHz')
Timing slack : 4763ps
Start-point  : dp/ir/data_out_reg[0]/CK
End-point    : data_bus[7]

```

Date-17/03/2015

ASIC Lab
Experiment No. 4

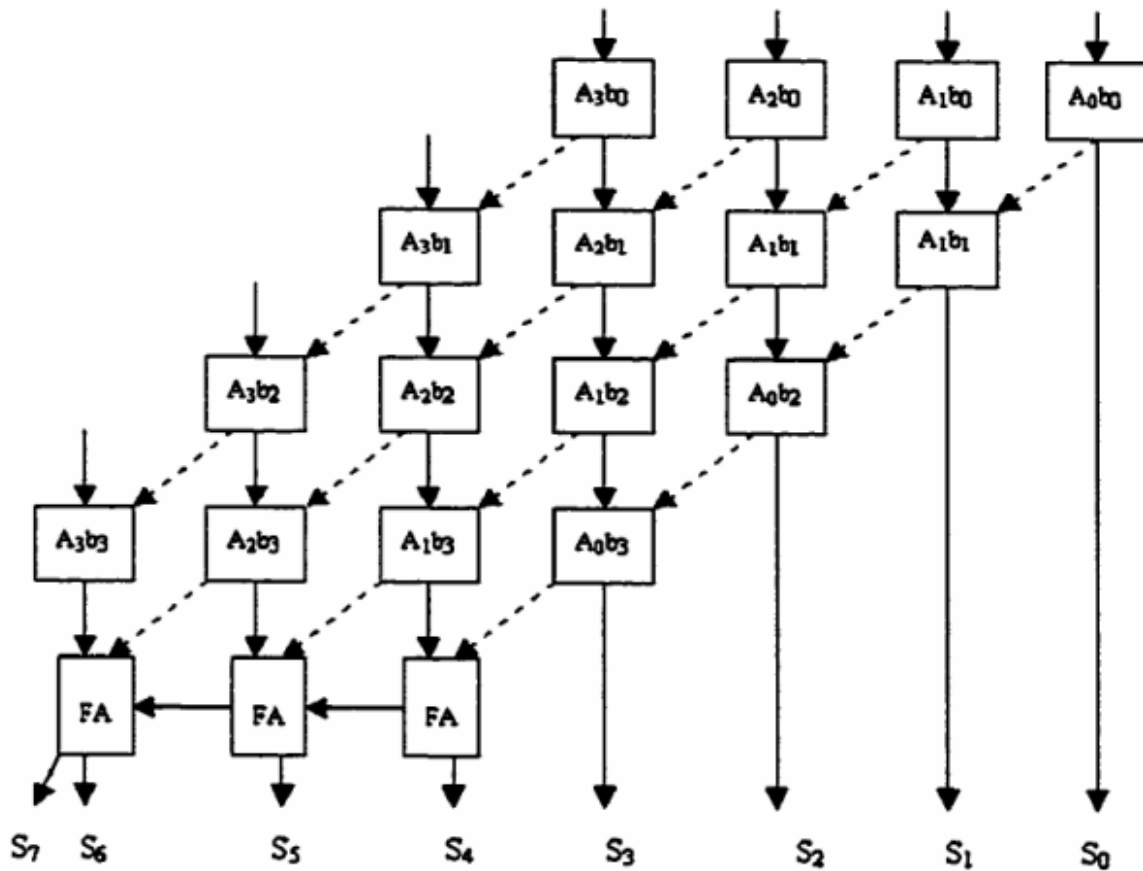
**Netlist and Sdc Generation of Multiplier
using RTL compiler**

**Baviskar Radheshyam
14MVD1039**

Aim: To generate Netlist and sdc timing file using rtl compiler.

Tools used: rc -64

Architectural Design:



HDL codes:

```
//----Half Adder-----//
```

```
module haad(a,b,s,c);  
input a,b;  
output c,s;
```

```
assign s= a^b;  
assign c= a&b;
```

```
endmodule
```

```
//----Full Adder-----//
```

```
module faa(a,b,cin,s,c);  
input a,b,cin ;  
output c,s ;  
  
assign s= a^b^cin;  
assign c= (a&b)|(b&cin)|(a&cin);  
  
endmodule
```

```
//-----Array Multiplier Rightshift----//
```

```
module shiftadd(a,b,c);  
input [3:0] a,b;  
output [7:0]c;  
wire [8:0]temp_car,temp_sum,t3,t4;  
  
assign c[0]=a[0]&b[0];  
haad ha1 ((a[1]&b[1]),(a[1]&b[0]),c[1],temp_car[0]);  
faa fa1 ((a[2]&b[0]),(a[1]&b[1]),(a[0]&b[2]),temp_sum[0],temp_car[1]);  
haad ha2 (temp_sum[0],temp_car[0],c[2],temp_car[2]);  
faa fa2 ((a[3]&b[0]),(a[2]&b[1]),(a[1]&b[2]),temp_sum[1],temp_car[3]);  
faa fa3 ((a[0]&b[3]),temp_sum[1],temp_car[3],temp_sum[2],temp_car[4]);  
faa fa4 (temp_sum[2],temp_car[4],temp_car[2],c[3],temp_car[5]); //add  
to next stage  
faa fa5 ((a[3]&b[1]),(a[2]&b[2]),(a[1]&b[3]),temp_sum[3],temp_car[6]);  
faa fa6 (temp_sum[3],temp_car[6],temp_car[5],c[4],temp_car[7]); //add  
to next stage  
faa fa7 ((a[3]&b[2]),(a[2]&b[3]),temp_car[7],c[5],temp_car[8]); //add  
to next  
  
haad ha3 ((a[3]&b[3]),temp_car[8],c[6],c[7]);  
  
endmodule
```

```
//-----Array Multiplier leftshift----//
```

```
module shiftaddl(a,b,c);  
input [3:0] a,b;  
output [7:0]c;  
wire [8:0]temp_car,temp_sum,t3,t4;  
  
assign c[7]=a[3]&b[3];
```

```

haad ha1 ((a[3]&b[2]),(a[2]&b[3]),c[6],temp_car[0]);
faa fa1 ((a[3]&b[1]),(a[2]&b[2]),(a[1]&b[3]),temp_sum[0],temp_car[1]);
haad ha2 (temp_sum[0],temp_car[0],temp_car[1],temp_car[2]);
faa fa2 ((a[3]&b[0]),(a[2]&b[1]),(a[1]&b[2]),temp_sum[1],temp_car[3]);
faa fa3 ((a[0]&b[3]),temp_sum[1],temp_car[3],temp_sum[2],temp_car[4]);
faa fa4 (temp_sum[2],temp_car[4],temp_car[2],c[3],temp_car[5]);//add
to next stage
faa fa5 ((a[3]&b[1]),(a[2]&b[2]),(a[1]&b[3]),temp_sum[3],temp_car[6]);
faa fa6 (temp_sum[3],temp_car[6],temp_car[5],c[4],temp_car[7]);//add
to next stage
faa fa7 ((a[3]&b[2]),(a[2]&b[3]),temp_car[7],c[5],temp_car[8]);//add
to next

haad ha3 ((a[3]&b[3]),temp_car[8],c[6],c[7]);

endmodule

```

SYNTHESIZED OUTPUT:

Reports:

Power

```

=====
Generated by:          Encounter(R) RTL Compiler v09.10-p104_1
Generated on:          Apr 27 2015  04:20:22 PM
Module:                shiftadd
Technology library:    tsmc18 1.0
Operating conditions:  slow (balanced_tree)
Wireload mode:         enclosed
Area mode:             timing library
=====

```

		Leakage	Dynamic	Total
Instance	Cells	Power (nW)	Power (nW)	Power (nW)
shiftadd	50	18.639	20769.183	20787.822
ha1	1	1.424	777.463	778.887

ha2	1	1.424	920.481	921.905
ha3	1	1.424	907.286	908.710
fa2	5	0.929	1796.300	1797.228
fa3	5	0.929	1835.683	1836.612
fa4	5	0.929	1506.960	1507.888
fa5	5	0.929	1952.002	1952.931
fa6	5	0.929	1267.054	1267.983
fa7	5	0.929	1161.866	1162.794
fa1	2	0.707	843.475	844.182

Gates

```

=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Apr 27 2015  04:20:22 PM
Module:           shiftadd
Technology library: tsmc18 1.0
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====

```

Gate	Instances	Area	Library
ADDHXL	3	109.771	tsmc18
AND2X1	15	199.584	tsmc18
INVXL	6	39.917	tsmc18
NAND2XL	6	59.875	tsmc18
NOR2XL	7	69.854	tsmc18
OAI21XL	12	159.667	tsmc18
OAI2BB2XL	1	23.285	tsmc18
total	50	661.954	

Type	Instances	Area	Area %
inverter	6	39.917	6.0
logic	44	622.037	94.0
total	50	661.954	100.0

Area

```

=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Apr 27 2015  04:25:48 PM
Module:           shiftaddl

```


Technology library: tsmc18 1.0
 Operating conditions: slow (balanced_tree)
 Wireload mode: enclosed
 Area mode: timing library

=====

Instance	Cells	Cell Area	Net Area	Wireload

shiftaddl	44	552	0	<none> (D)
fa7	5	53	0	<none> (D)
fa6	5	53	0	<none> (D)
fa4	5	53	0	<none> (D)
fa3	5	53	0	<none> (D)
fa2	5	53	0	<none> (D)
fa1	5	53	0	<none> (D)
ha3	1	37	0	<none> (D)
ha1	1	37	0	<none> (D)
ha2	1	13	0	<none> (D)

=====

NETLIST

```
// Generated by Cadence Encounter(R) RTL Compiler v09.10-p104_1

module faa(a, b, cin, s, c);
  input a, b, cin;
  output s, c;
  wire a, b, cin;
  wire s, c;
  wire n_0;
  assign c = 1'b0;
  OAI2BB2XL g37(.A0N (b), .A1N (a), .B0 (n_0), .B1 (cin), .Y (s));
  NOR2XL g38(.A (a), .B (b), .Y (n_0));
endmodule

module faa_1(a, b, cin, s, c);
  input a, b, cin;
  output s, c;
  wire a, b, cin;
  wire s, c;
  wire n_0, n_1, n_2;
  OAI21XL g72(.A0 (n_1), .A1 (cin), .B0 (n_2), .Y (s));
  OAI21XL g73(.A0 (n_1), .A1 (n_0), .B0 (n_2), .Y (c));
endmodule
```

```

    NAND2XL g74(.A (b), .B (a), .Y (n_2));
    NOR2XL g75(.A (b), .B (a), .Y (n_1));
    INVXL g76(.A (cin), .Y (n_0));
endmodule

module faa_2(a, b, cin, s, c);
    input a, b, cin;
    output s, c;
    wire a, b, cin;
    wire s, c;
    wire n_0, n_1, n_2;
    OAI21XL g72(.A0 (n_1), .A1 (cin), .B0 (n_2), .Y (s));
    OAI21XL g73(.A0 (n_1), .A1 (n_0), .B0 (n_2), .Y (c));
    NAND2XL g74(.A (a), .B (b), .Y (n_2));
    NOR2XL g75(.A (b), .B (a), .Y (n_1));
    INVXL g76(.A (cin), .Y (n_0));
endmodule

module faa_3(a, b, cin, s, c);
    input a, b, cin;
    output s, c;
    wire a, b, cin;
    wire s, c;
    wire n_0, n_1, n_2;
    OAI21XL g72(.A0 (n_1), .A1 (cin), .B0 (n_2), .Y (s));
    OAI21XL g73(.A0 (n_1), .A1 (n_0), .B0 (n_2), .Y (c));
    NAND2XL g74(.A (b), .B (a), .Y (n_2));
    NOR2XL g75(.A (a), .B (b), .Y (n_1));
    INVXL g76(.A (cin), .Y (n_0));
endmodule

module faa_4(a, b, cin, s, c);
    input a, b, cin;
    output s, c;
    wire a, b, cin;
    wire s, c;
    wire n_0, n_1, n_2;
    OAI21XL g72(.A0 (n_1), .A1 (cin), .B0 (n_2), .Y (s));
    OAI21XL g73(.A0 (n_1), .A1 (n_0), .B0 (n_2), .Y (c));
    NAND2XL g74(.A (b), .B (a), .Y (n_2));
    NOR2XL g75(.A (b), .B (a), .Y (n_1));
    INVXL g76(.A (cin), .Y (n_0));
endmodule

module faa_5(a, b, cin, s, c);
    input a, b, cin;
    output s, c;
    wire a, b, cin;
    wire s, c;
    wire n_0, n_1, n_2;
    OAI21XL g72(.A0 (cin), .A1 (n_1), .B0 (n_2), .Y (s));
    OAI21XL g73(.A0 (n_0), .A1 (n_1), .B0 (n_2), .Y (c));

```

```

    NAND2XL g74(.A (b), .B (a), .Y (n_2));
    NOR2XL g75(.A (a), .B (b), .Y (n_1));
    INVXL g76(.A (cin), .Y (n_0));
endmodule

module faa_6(a, b, cin, s, c);
    input a, b, cin;
    output s, c;
    wire a, b, cin;
    wire s, c;
    wire n_0, n_1, n_2;
    OAI21XL g72(.A0 (cin), .A1 (n_1), .B0 (n_2), .Y (s));
    OAI21XL g73(.A0 (n_0), .A1 (n_1), .B0 (n_2), .Y (c));
    NAND2XL g74(.A (b), .B (a), .Y (n_2));
    NOR2XL g75(.A (b), .B (a), .Y (n_1));
    INVXL g76(.A (cin), .Y (n_0));
endmodule

module haad(a, b, s, c);
    input a, b;
    output s, c;
    wire a, b;
    wire s, c;
    ADDHXL g17(.A (a), .B (b), .S (s), .CO (c));
endmodule

module haad_1(a, b, s, c);
    input a, b;
    output s, c;
    wire a, b;
    wire s, c;
    ADDHXL g17(.A (b), .B (a), .S (s), .CO (c));
endmodule

module haad_2(a, b, s, c);
    input a, b;
    output s, c;
    wire a, b;
    wire s, c;
    ADDHXL g17(.A (a), .B (b), .S (s), .CO (c));
endmodule

module shiftadd(a, b, c);
    input [3:0] a, b;
    output [7:0] c;
    wire [3:0] a, b;
    wire [7:0] c;
    wire [8:0] temp_sum;
    wire [8:0] temp_car;
    wire UNCONNECTED, n_0, n_1, n_2, n_3, n_4, n_7, n_18;
    wire n_23, n_24, n_25, n_26, n_27, n_28, n_29;
    faa fa1(n_7, n_29, n_4, temp_sum[0], UNCONNECTED);

```

```

faa_1 fa2(n_3, n_2, n_1, temp_sum[1], temp_car[3]);
faa_2 fa3(n_0, temp_sum[1], temp_car[3], temp_sum[2], temp_car[4]);
faa_3 fa4(temp_sum[2], temp_car[4], temp_car[2], c[3], temp_car[5]);
faa_4 fa5(n_28, n_27, n_26, temp_sum[3], temp_car[6]);
faa_5 fa6(temp_sum[3], temp_car[6], temp_car[5], c[4], temp_car[7]);
faa_6 fa7(n_25, n_24, temp_car[7], c[5], temp_car[8]);
haad ha1(n_29, n_18, c[1], temp_car[0]);
haad_1 ha2(temp_sum[0], temp_car[0], c[2], temp_car[2]);
haad_2 ha3(n_23, temp_car[8], c[6], c[7]);
AND2X1 g107(.A (a[0]), .B (b[0]), .Y (c[0]));
AND2X1 g108(.A (a[0]), .B (b[3]), .Y (n_0));
AND2X1 g109(.A (a[1]), .B (b[2]), .Y (n_1));
AND2X1 g110(.A (a[2]), .B (b[2]), .Y (n_27));
AND2X1 g111(.A (a[3]), .B (b[3]), .Y (n_23));
AND2X1 g112(.A (a[1]), .B (b[1]), .Y (n_29));
AND2X1 g113(.A (a[1]), .B (b[0]), .Y (n_18));
AND2X1 g114(.A (a[2]), .B (b[0]), .Y (n_7));
AND2X1 g115(.A (a[0]), .B (b[2]), .Y (n_4));
AND2X1 g116(.A (a[3]), .B (b[0]), .Y (n_3));
AND2X1 g117(.A (a[2]), .B (b[3]), .Y (n_24));
AND2X1 g118(.A (a[2]), .B (b[1]), .Y (n_2));
AND2X1 g119(.A (a[3]), .B (b[1]), .Y (n_28));
AND2X1 g120(.A (a[1]), .B (b[3]), .Y (n_26));
AND2X1 g121(.A (a[3]), .B (b[2]), .Y (n_25));
endmodule

```

sdc file

```

# #####

# Created by Encounter(R) RTL Compiler v09.10-p104_1 on Mon Apr 27
16:20:22 IST 2015

# #####

set_sdc_version 1.7

set_units -capacitance 1000.0fF
set_units -time 1000.0ps

# Set the current design
current_design shiftadd

set_clock_gating_check -setup 0.0
set_wire_load_mode "enclosed"
set_dont_use [get_lib_cells tsmc18/RF1R1WX2]
set_dont_use [get_lib_cells tsmc18/RF2R1WX2]

```

```
set_dont_use [get_lib_cells tsmc18/RFRDX1]  
set_dont_use [get_lib_cells tsmc18/RFRDX2]  
set_dont_use [get_lib_cells tsmc18/RFRDX4]  
set_dont_use [get_lib_cells tsmc18/TIEHI]  
set_dont_use [get_lib_cells tsmc18/TIELO]
```

INFERENCE

Using RTL compiler Netlist and sdc file is generated for the array Multiplier.

Date-24/03/2015

ASIC Lab
Experiment No. 5

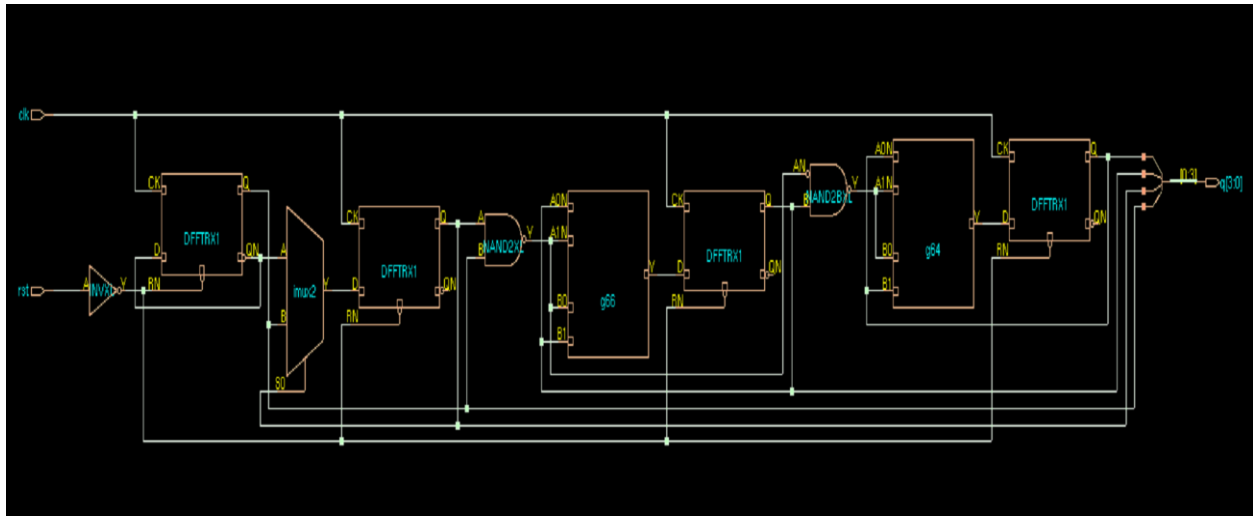
**LOW POWER OPTIMIZATION OF
COUNTER USING RC COMPILER**

**Baviskar Radheshyam
14MVD1039**

AIM: To design and synthesize 4-bit counter and perform low power optimization.

TOOLS USED: Cadence RTL Compiler 64-bit

RTL SCHEMATIC:



VERILOG CODE:

```
module counter4(clk,rst,q);
input clk,rst;
output reg [3:0] q;
always @(posedge clk)
begin
if(rst==1'b1)
q<=4'b0;
else
q<=q+1'b1;
end
endmodule
```

TESTBENCH:

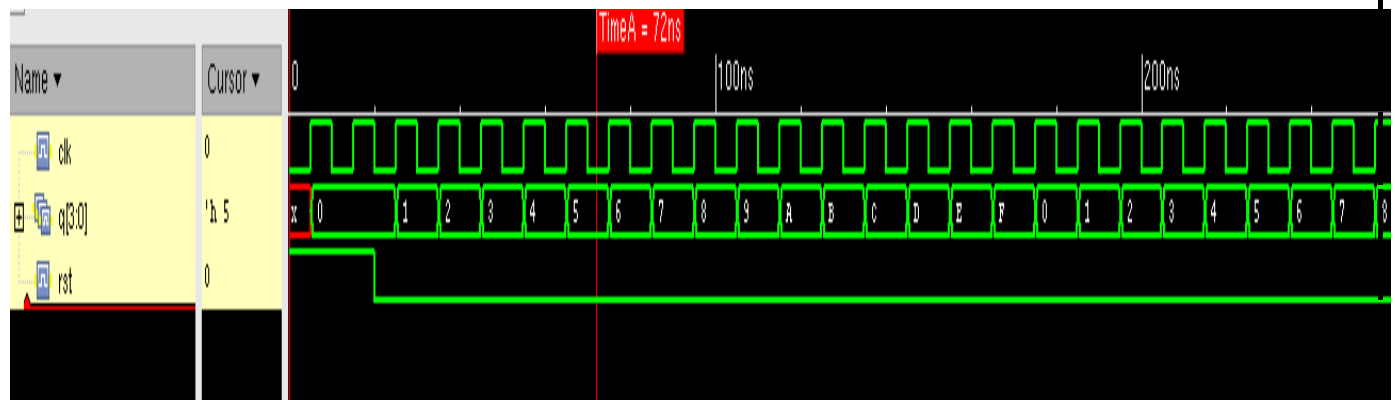
```
module test_counter4();
reg clk=1'b0;
reg rst;
wire [3:0] q;
counter4 z1(clk,rst,q);
always #5 clk=~clk;
initial
begin
rst=1'b1;
```

```

#20 rst=1'b0;
end
initial
begin
$monitor("clk=%b,rst=%b,q=%b",clk,rst,q);
#400 $finish;
end
endmodule

```

SIMULATION RESULT:



```

set_attribute source_verbose true
set_attribute information_level 9
set_attribute library library/slow.lib
read_hdl rtl/counter_rtl.v
elaborate
synthesize -to_mapped -effort low
report area > area.txt
report power > power.txt
report timing > timing.txt
report gates > gates.txt
exit

```

```

=====
Generated by:          Encounter(R) RTL Compiler v09.10-p104_1
Generated on:          Mar 24 2015  11:30:52 AM
Module:                counter4
Technology library:    tsmc18 1.0
Operating conditions:  slow (balanced_tree)
Wireload mode:         enclosed
Area mode:             timing library
=====

```

Instance	Cells	Cell Area	Net Area	Wireload
counter4	11	336	0	<none> (D)

inc_add_12_5_1 5 96 0 <none> (D)

(D) = wireload is default in technology library

```
=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Mar 24 2015  11:30:52 AM
Module:           counter4
Technology library: tsmc18 1.0
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====
```

Gate	Instances	Area	Library
DFFHQX1	1	53.222	tsmc18
DFFTRX1	3	169.646	tsmc18
INVX1	1	6.653	tsmc18
NAND2BXL	1	13.306	tsmc18
NAND2XL	1	9.979	tsmc18
NOR2XL	1	9.979	tsmc18
OAI2BB2XL	2	46.570	tsmc18
XOR2XL	1	26.611	tsmc18
total	11	335.966	

Type	Instances	Area	Area %
sequential	4	222.869	66.3
inverter	1	6.653	2.0
logic	6	106.445	31.7
total	11	335.966	100.0

```
=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Mar 24 2015  11:30:52 AM
Module:           counter4
Technology library: tsmc18 1.0
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====
```

Instance	Cells	Leakage Power (nW)	Dynamic Power (nW)	Total Power (nW)
----------	-------	--------------------	--------------------	------------------

```

-----
counter4          11      9.990  8759.463  8769.453
  inc_add_12_5_1    5      3.257  1118.050  1121.307

```

```

=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Mar 24 2015  11:30:52 AM
Module:           counter4
Technology library: tsmc18 1.0
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====

```

Pin	Type	Fanout	Load (fF)	Slew (ps)	Delay (ps)	Arrival (ps)	
q_reg[1]/CK				0		0	R
q_reg[1]/Q	DDFTRX1	3	5.2	122	+367	367	R
inc_add_12_5_1/A[1]							
g25/A					+0	367	
g25/Y	NAND2XL	3	7.6	134	+105	473	F
g23/AN					+0	473	
g23/Y	NAND2BXL	2	5.5	118	+204	677	F
g21/A1N					+0	677	
g21/Y	OAI2BB2XL	1	2.2	105	+193	870	F
inc_add_12_5_1/Z[3]							
q_reg[3]/D	DDFTRX1				+0	870	
q_reg[3]/CK	setup			0	+410	1280	R

```

-----
Timing slack : UNCONSTRAINED
Start-point  : q_reg[1]/CK
End-point    : q_reg[3]/D

```

```

-----
set_attribute source_verbose true
set_attribute information_level 9
set_attribute library library/slow.lib
read_hdl rtl/counter_rtl.v
elaborate
set_attribute lp_insert_clock_gating true /
synthesize -to_mapped -effort low
report area > area1.txt
report power > power1.txt
report timing > timing1.txt
report gates > gates1.txt
exit
-----
=====

```

Generated by: Encounter(R) RTL Compiler v09.10-p104_1
 Generated on: Mar 24 2015 11:32:17 AM
 Module: counter4
 Technology library: tsmc18 1.0
 Operating conditions: slow (balanced_tree)
 Wireload mode: enclosed
 Area mode: timing library

Instance	Cells	Cell Area	Net Area	Wireload
counter4	14	406	0	<none> (D)
RC_CG_HIER_INST1	3	67	0	<none> (D)

(D) = wireload is default in technology library

Generated by: Encounter(R) RTL Compiler v09.10-p104_1
 Generated on: Mar 24 2015 11:32:17 AM
 Module: counter4
 Technology library: tsmc18 1.0
 Operating conditions: slow (balanced_tree)
 Wireload mode: enclosed
 Area mode: timing library

Gate	Instances	Area	Library
AND2X1	1	13.306	tsmc18
DFFTRX1	4	226.195	tsmc18
INVX1	1	6.653	tsmc18
MXI2XL	1	23.285	tsmc18
NAND2BXL	1	13.306	tsmc18
NAND2XL	1	9.979	tsmc18
OAI2BB2XL	2	46.570	tsmc18
OR2X1	1	13.306	tsmc18
OR3XL	1	16.632	tsmc18
TLATNX1	1	36.590	tsmc18
total	14	405.821	

Type	Instances	Area	Area %
sequential	5	262.786	64.8
inverter	1	6.653	1.6
logic	8	136.382	33.6
total	14	405.821	100.0

```

=====
Generated by:          Encounter(R) RTL Compiler v09.10-p104_1
Generated on:          Mar 24 2015  11:32:17 AM
Module:                counter4
Technology library:    tsmc18 1.0
Operating conditions:  slow (balanced_tree)
Wireload mode:         enclosed
Area mode:             timing library
=====

```

Instance	Cells	Leakage Power (nW)	Dynamic Power (nW)	Total Power (nW)
counter4	14	13.016	9518.732	9531.748
RC_CG_HIER_INST1	3	2.955	2005.631	2008.586

```

=====
Generated by:          Encounter(R) RTL Compiler v09.10-p104_1
Generated on:          Mar 24 2015  11:32:17 AM
Module:                counter4
Technology library:    tsmc18 1.0
Operating conditions:  slow (balanced_tree)
Wireload mode:         enclosed
Area mode:             timing library
=====

```

Pin	Type	Fanout	Load (fF)	Slew (ps)	Delay (ps)	Arrival (ps)	
q_reg[1]/CK				0		0	R
q_reg[1]/Q	DFFTRX1	3	8.9	165	+394	394	R
g58/B					+0	394	
g58/Y	NAND2XL	3	7.6	132	+113	507	F
g55/AN					+0	507	
g55/Y	NAND2BXL	2	5.5	118	+203	710	F
g52/A1N					+0	710	
g52/Y	OAI2BB2XL	1	1.8	101	+190	900	F
q_reg[3]/RN	DFFTRX1				+0	900	
q_reg[3]/CK	setup			0	+437	1337	R

```

Timing slack : UNCONSTRAINED
Start-point  : q_reg[1]/CK
End-point    : q_reg[3]/RN

```

```

-----
set_attribute source_verbose true
set_attribute information_level 9
set_attribute library library/slow.lib
read_hdl rtl/counter_rtl.v

```

```

elaborate
set_attribute lp_insert_operand_isolation true /
synthesize -to_mapped -effort low
report area > area2.txt
report power > power2.txt
report timing > timing2.txt
report gates > gates2.txt
exit

```

```

=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Mar 24 2015  11:33:47 AM
Module:            counter4
Technology library: tsmc18 1.0
Operating conditions: slow (balanced_tree)
Wireload mode:     enclosed
Area mode:         timing library
=====

```

Instance	Cells	Cell Area	Net Area	Wireload
counter4	11	336	0	<none> (D)
inc_add_12_5_1	5	96	0	<none> (D)

(D) = wireload is default in technology library

```

=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Mar 24 2015  11:33:47 AM
Module:            counter4
Technology library: tsmc18 1.0
Operating conditions: slow (balanced_tree)
Wireload mode:     enclosed
Area mode:         timing library
=====

```

Gate	Instances	Area	Library
DFFHQX1	1	53.222	tsmc18
DFFTRX1	3	169.646	tsmc18
INVX1	1	6.653	tsmc18
NAND2BXL	1	13.306	tsmc18
NAND2XL	1	9.979	tsmc18
NOR2XL	1	9.979	tsmc18
OAI2BB2XL	2	46.570	tsmc18
XOR2XL	1	26.611	tsmc18
total	11	335.966	

Type	Instances	Area	Area %
sequential	4	222.869	66.3
inverter	1	6.653	2.0
logic	6	106.445	31.7
total	11	335.966	100.0

```

=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Mar 24 2015  11:33:47 AM
Module:            counter4
Technology library: tsmc18 1.0
Operating conditions: slow (balanced_tree)
Wireload mode:     enclosed
Area mode:         timing library
=====

```

Instance	Cells	Leakage Power (nW)	Dynamic Power (nW)	Total Power (nW)
counter4	11	9.990	8759.463	8769.453
inc_add_12_5_1	5	3.257	1118.050	1121.307

```

=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Mar 24 2015  11:33:47 AM
Module:            counter4
Technology library: tsmc18 1.0
Operating conditions: slow (balanced_tree)
Wireload mode:     enclosed
Area mode:         timing library
=====

```

Pin	Type	Fanout	Load (fF)	Slew (ps)	Delay (ps)	Arrival (ps)	
q_reg[1]/CK				0		0	R
q_reg[1]/Q	DDFTRX1	3	5.2	122	+367	367	R
inc_add_12_5_1/A[1]							
g25/A					+0	367	
g25/Y	NAND2XL	3	7.6	134	+105	473	F
g23/AN					+0	473	
g23/Y	NAND2BXL	2	5.5	118	+204	677	F
g21/A1N					+0	677	
g21/Y	OAI2BB2XL	1	2.2	105	+193	870	F
inc_add_12_5_1/Z[3]							
q_reg[3]/D	DDFTRX1				+0	870	
q_reg[3]/CK	setup			0	+410	1280	R

```
-----
Timing slack : UNCONSTRAINED
Start-point  : q_reg[1]/CK
End-point    : q_reg[3]/D
-----
```

```
set_attribute source_verbose true
set_attribute information_level 9
set_attribute library library/slow.lib
read_hdl rtl/counter_rtl.v
elaborate
set_attribute lp_operand_isolation_prefix string /
synthesize -to_mapped -effort low
report area > area3.txt
report power > power3.txt
report timing > timing3.txt
report gates > gates3.txt
exit
```

```
=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Mar 24 2015  11:34:05 AM
Module:           counter4
Technology library: tsmc18 1.0
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====
```

Instance	Cells	Cell Area	Net Area	Wireload
counter4	11	336	0	<none> (D)
inc_add_12_5_1	5	96	0	<none> (D)

(D) = wireload is default in technology library

```
=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Mar 24 2015  11:34:05 AM
Module:           counter4
Technology library: tsmc18 1.0
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====
```

Gate	Instances	Area	Library
------	-----------	------	---------

DFFHQX1	1	53.222	tsmc18
DFFTRX1	3	169.646	tsmc18
INVX1	1	6.653	tsmc18
NAND2BXL	1	13.306	tsmc18
NAND2XL	1	9.979	tsmc18
NOR2XL	1	9.979	tsmc18
OAI2BB2XL	2	46.570	tsmc18
XOR2XL	1	26.611	tsmc18

total	11	335.966	

Type	Instances	Area	Area %

sequential	4	222.869	66.3
inverter	1	6.653	2.0
logic	6	106.445	31.7

total	11	335.966	100.0

```

=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Mar 24 2015  11:34:05 AM
Module:           counter4
Technology library: tsmc18 1.0
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====

```

Instance	Cells	Leakage Power (nW)	Dynamic Power (nW)	Total Power (nW)

counter4	11	9.990	8759.463	8769.453
inc_add_12_5_1	5	3.257	1118.050	1121.307

```

=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Mar 24 2015  11:34:05 AM
Module:           counter4
Technology library: tsmc18 1.0
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====

```

Pin	Type	Fanout	Load (fF)	Slew (ps)	Delay (ps)	Arrival (ps)

q_reg[1]/CK			0		0	R


```

q_reg[1]/Q      DFFTRX1          3  5.2  122  +367      367 R
inc_add_12_5_1/A[1]
g25/A          +0      367
g25/Y          NAND2XL          3  7.6  134  +105      473 F
g23/AN          +0      473
g23/Y          NAND2BXL         2  5.5  118  +204      677 F
g21/A1N          +0      677
g21/Y          OAI2BB2XL        1  2.2  105  +193      870 F
inc_add_12_5_1/Z[3]
q_reg[3]/D      DFFTRX1          +0      870
q_reg[3]/CK      setup          0  +410     1280 R
-----

```

```

Timing slack : UNCONSTRAINED
Start-point  : q_reg[1]/CK
End-point    : q_reg[3]/D

```

COMPARISION TABLE:

PARAMETER	TYPICAL	CLOCK GATING	OPERAND ISOLATION	OPERAND ISOLATION PREFIX
AREA	336	406	336	336
POWER	8769.453	9531.748	8769.453	8769.453
GATES	11	14	11	11

INFERENCE: Hence, we have designed and synthesized 4-bit counter and did comparative analysis for the low power optimization approaches.

Date-21/04/2015

ASIC Lab
Experiment No. 6

**Layout Generation of Counter using
Encounter tool**

**Baviskar Radheshyam
14MVD1039**

Aim: To generate Layout of counter using encounter tool

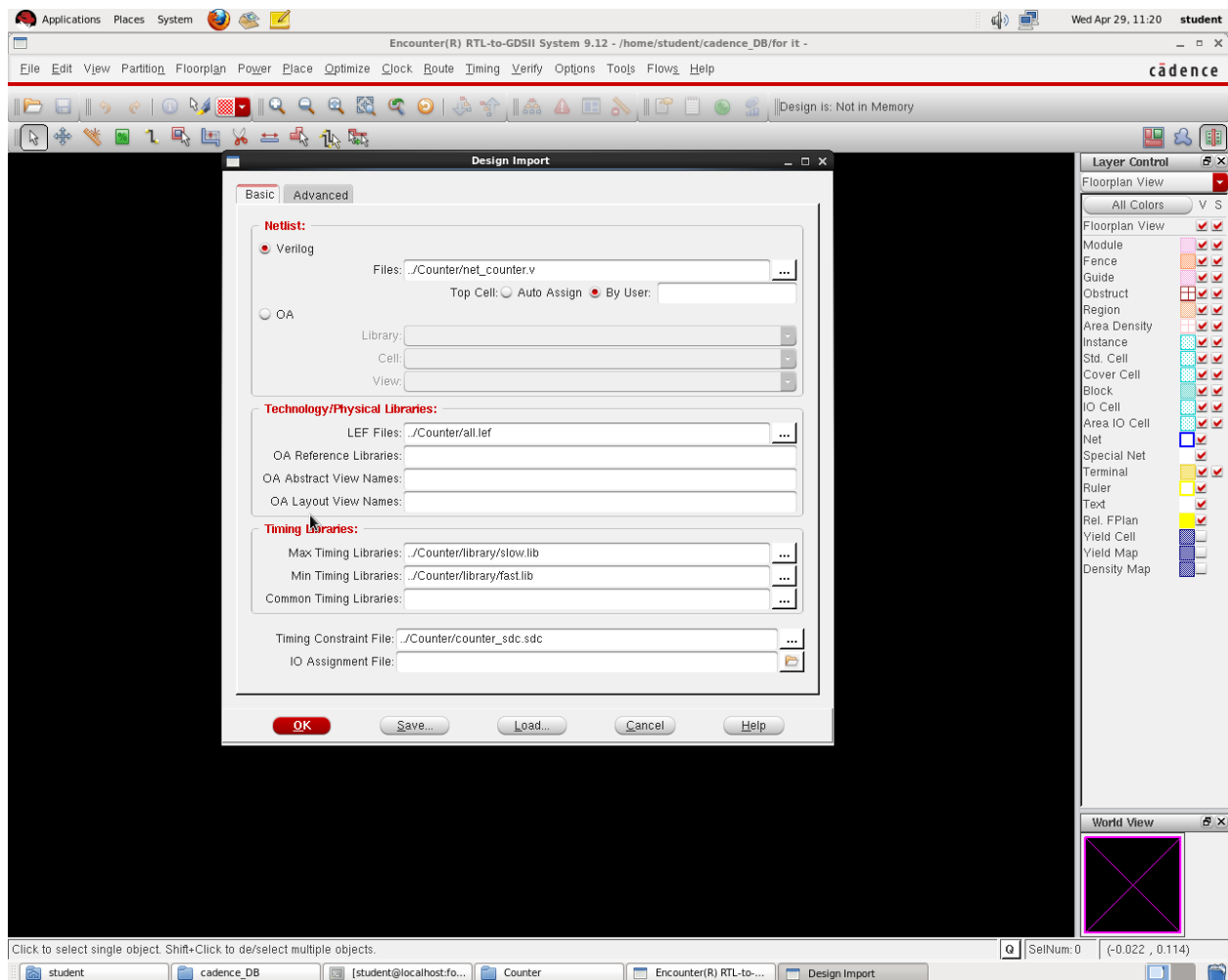
Tools used: cadence encounter

STEPS :

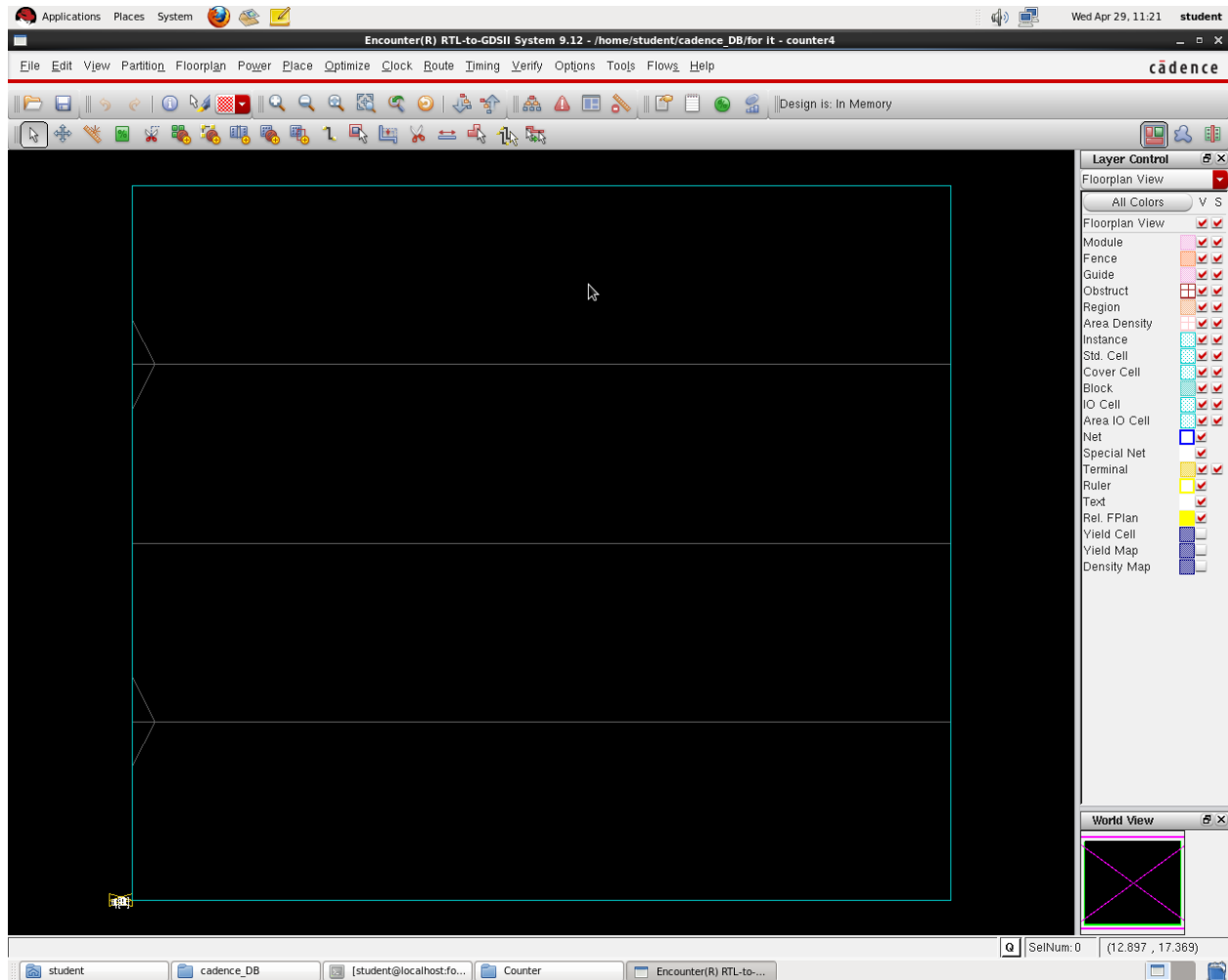
1.Requirements-

- a) Netlist file of the file
- b) sdc file

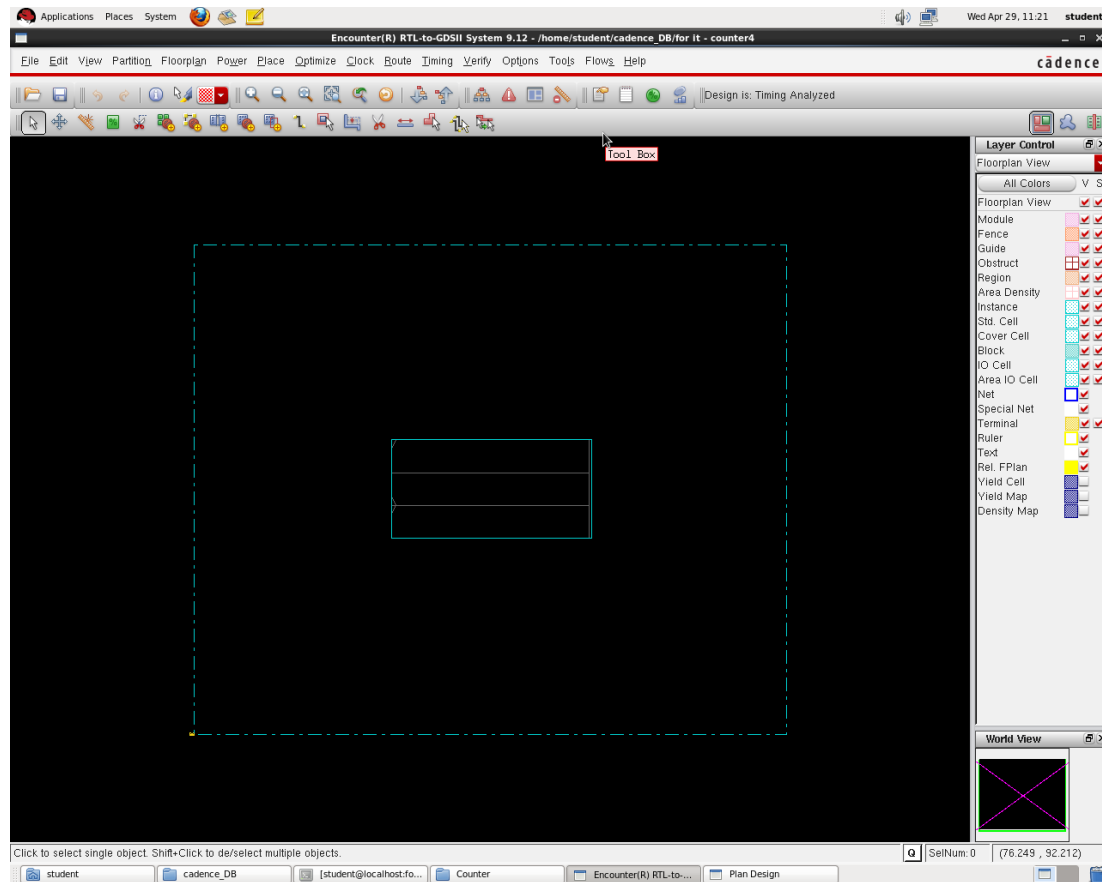
2.Import Design-



- Select Auto Design
- Give all.lef file
- for maximum select slow.lib
- for Minimum select fast.lib
- In Advance tab give Power as VDD and VSS



3.select specify floorplan



--Give all Design as 30um boundry

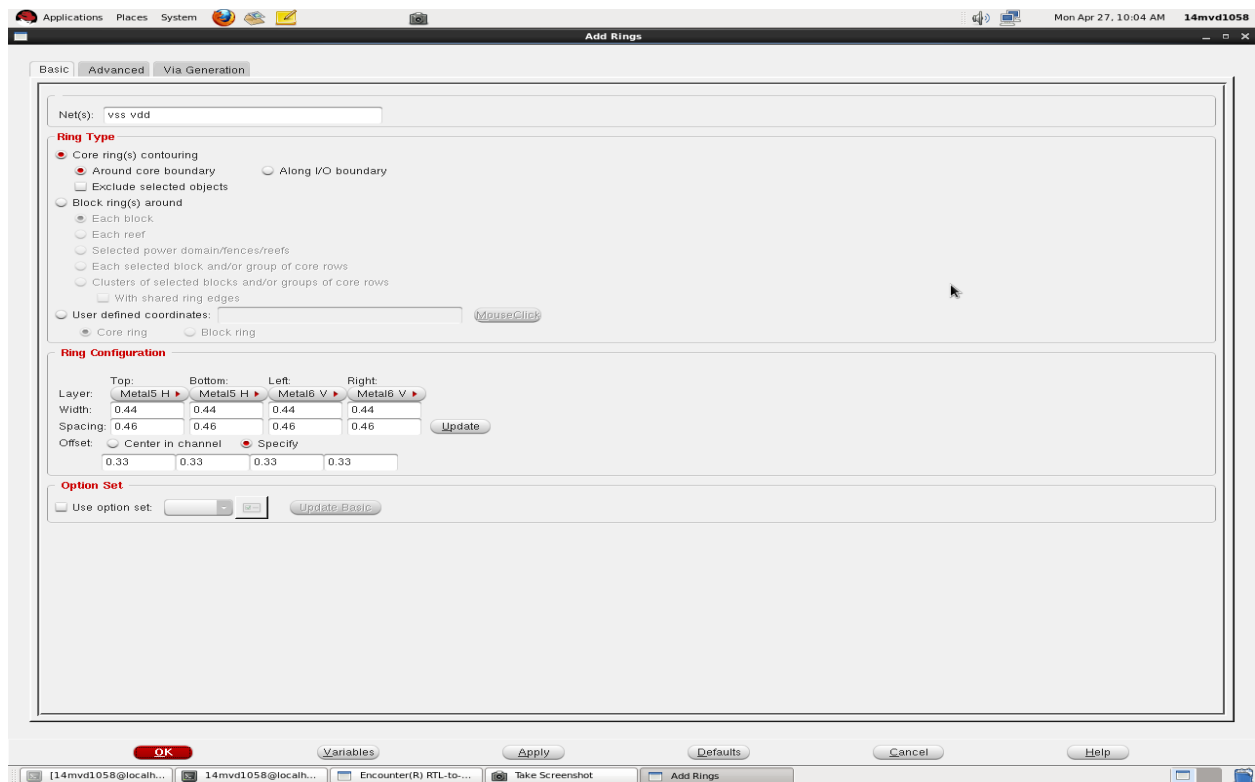
4. Select Power planning

For Add Strips-

--Give horizontal as metal 5

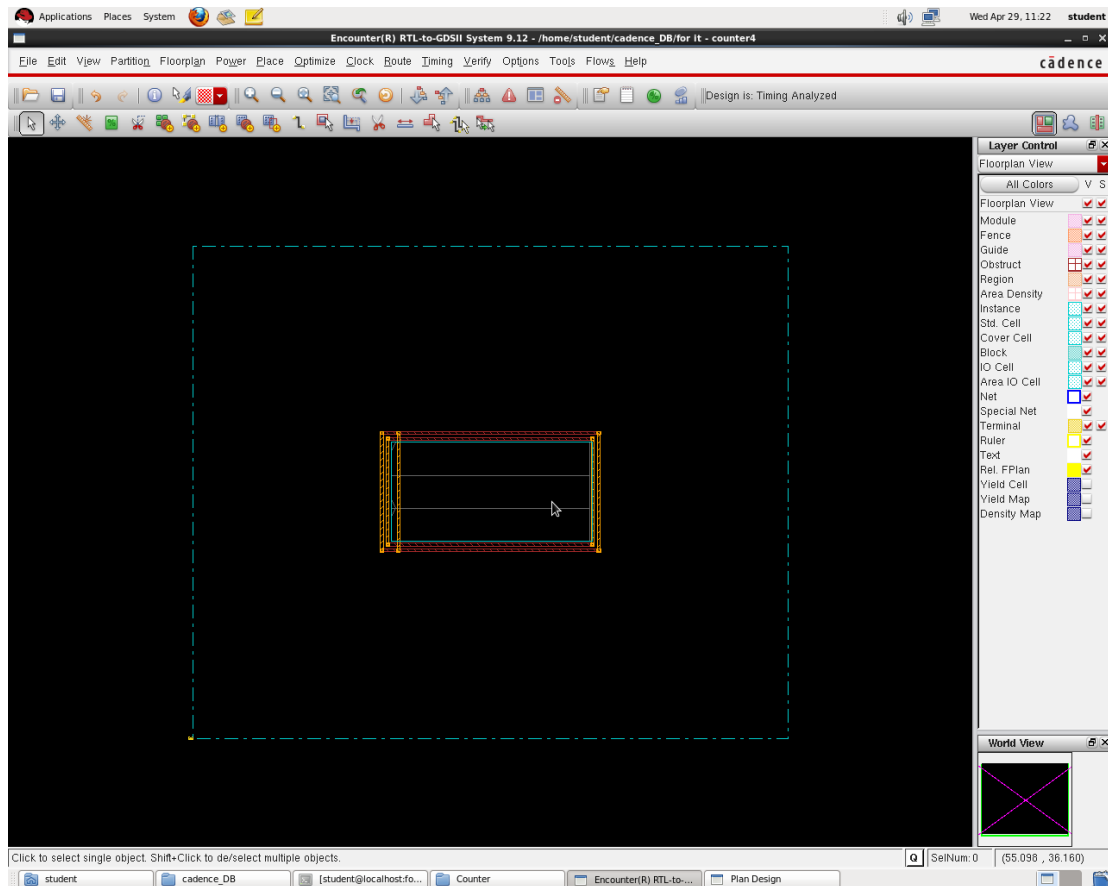
--Give Vertical as Metal 6

For Add Rings-



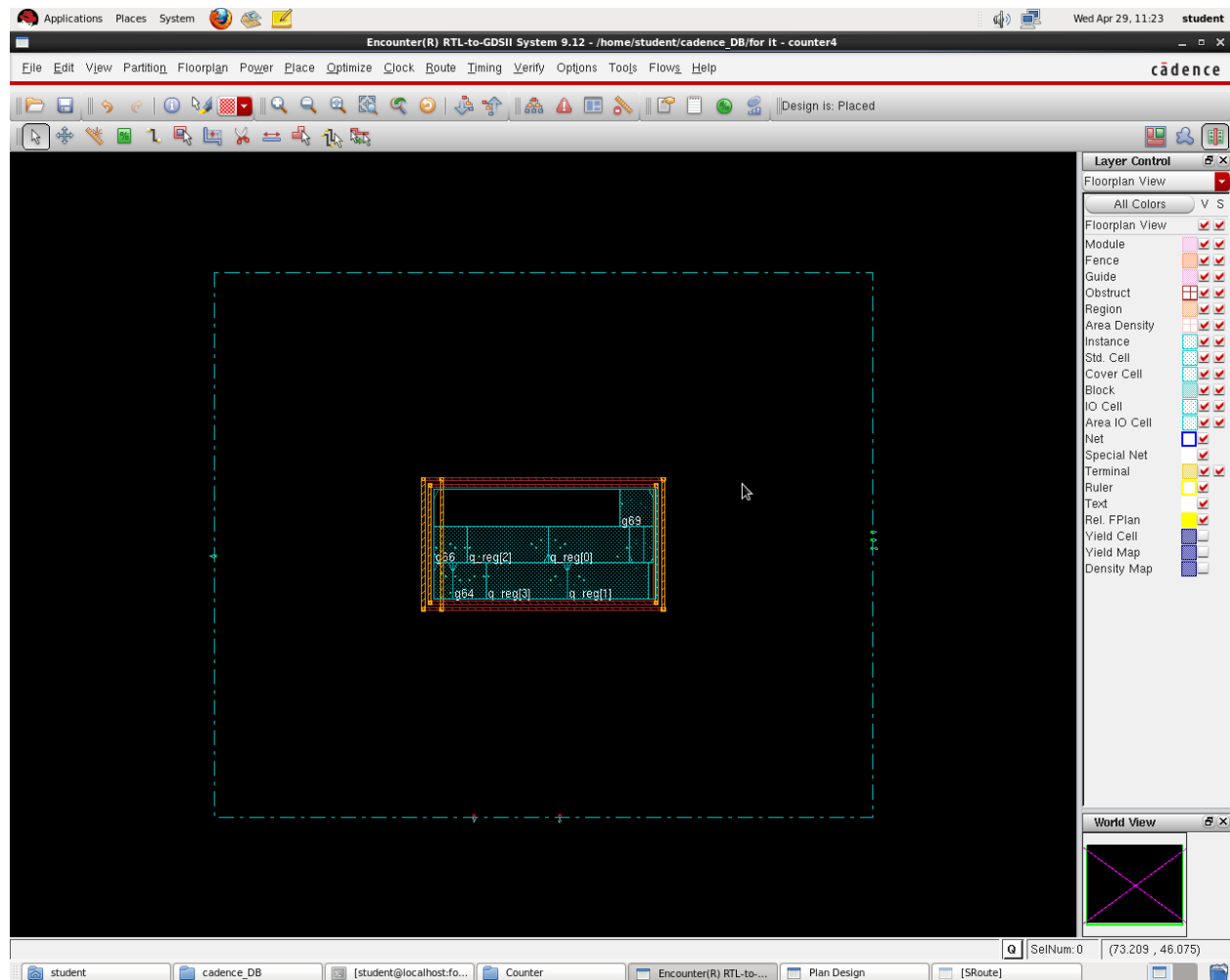
5.

Select the top and bottom layer as Metal5, Left and Right as Metal6. Set the width as per the requirement and taking the space between core boundary and I/O pad considerations. Select the option for offset as —center in channel and click OK. The power ring will get created in between the channel. The image on the next page is showing the power ring created.



6.

After the power planning, go to Route and click Special Route
Click OK with all default settings. This is done to provide power to standard cells.
The horizontal blue coloured metal1 stripes created as a result of Special Route

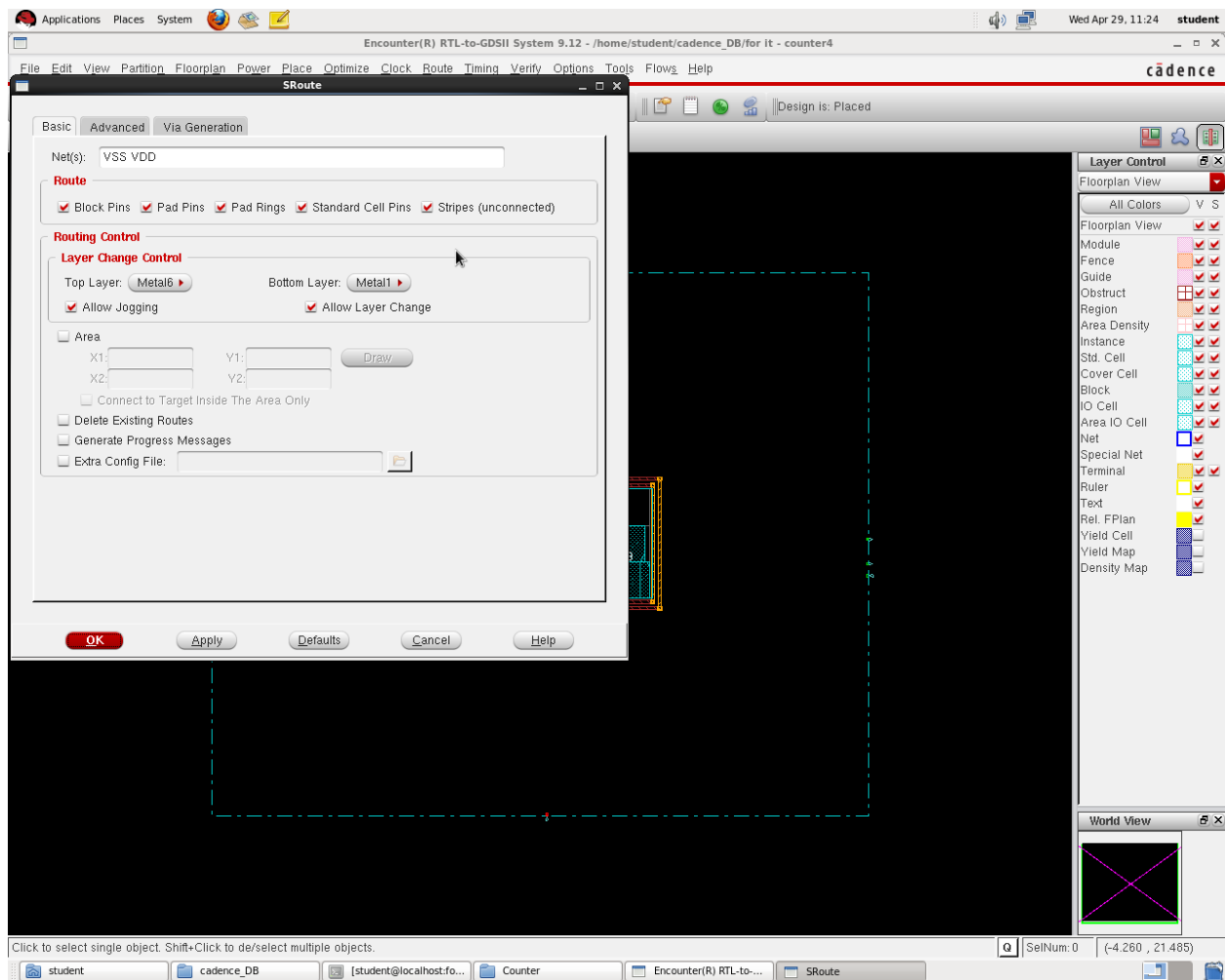


7

For placement, click on place and select place and click on Place Standard Cell.
Click OK on Place window and in physical view the blue coloured standard cells
can be seen as a result of placement of standard cells

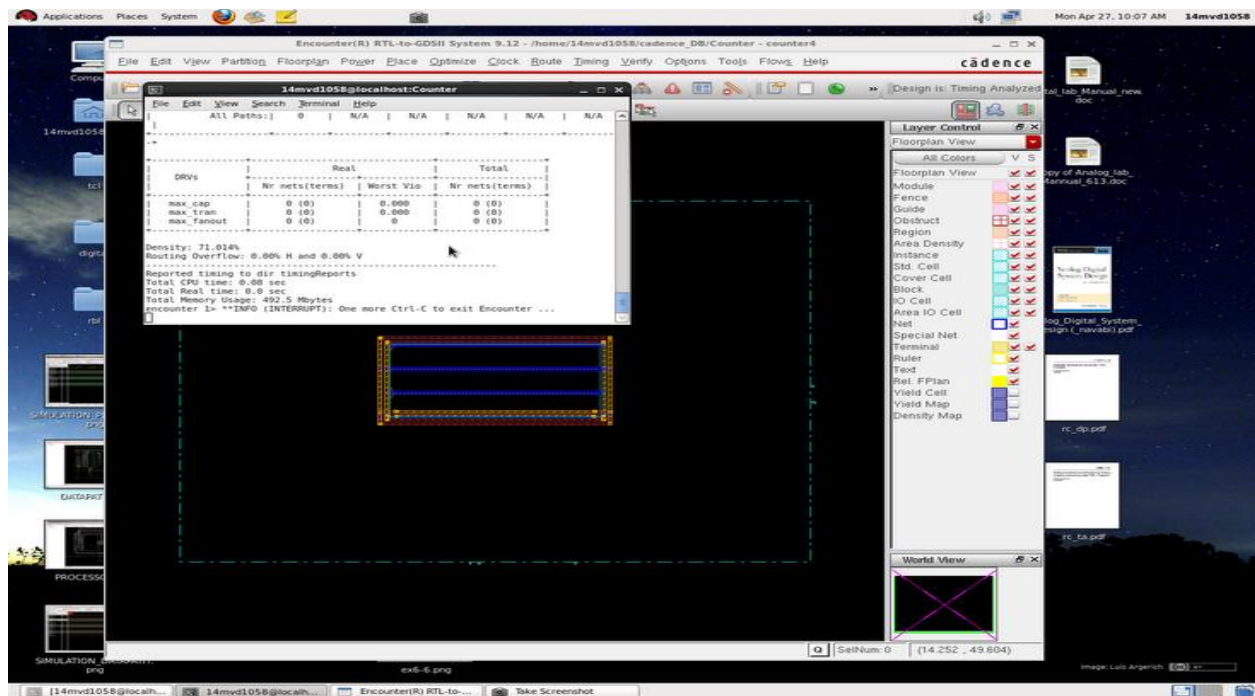
8

Before CTS, timing analysis has to be done for any setup violations. Click on
Timing, and select Report Timing. A Timing analysis window will get open. In the
window select the —Pre-CTS as Design Stage and select the —Setup as Analysis
Type



9.

Click OK to complete the Timing analysis. The timing information will get display on terminal in tabular form. In the table displayed on the terminal under —timeDesign Summary||, check for any negative value under WNS(Worst Negative Slack) and TNS(Total Negative Slack). The terminal will look as the image below and Tool window as on next page.



10.

If there is any of the negative slack value under WNS or TNS, click Optimize in Tool window and Select Optimize Design. A new window —Optimization will get open. Select —Pre-CTS as Design Stage and —Setup as optimization type and click OK. The tool will optimize the design and the optimized timing results will be displayed over terminal again.

In this case we did not get any negative slack, so this step is skipped here.

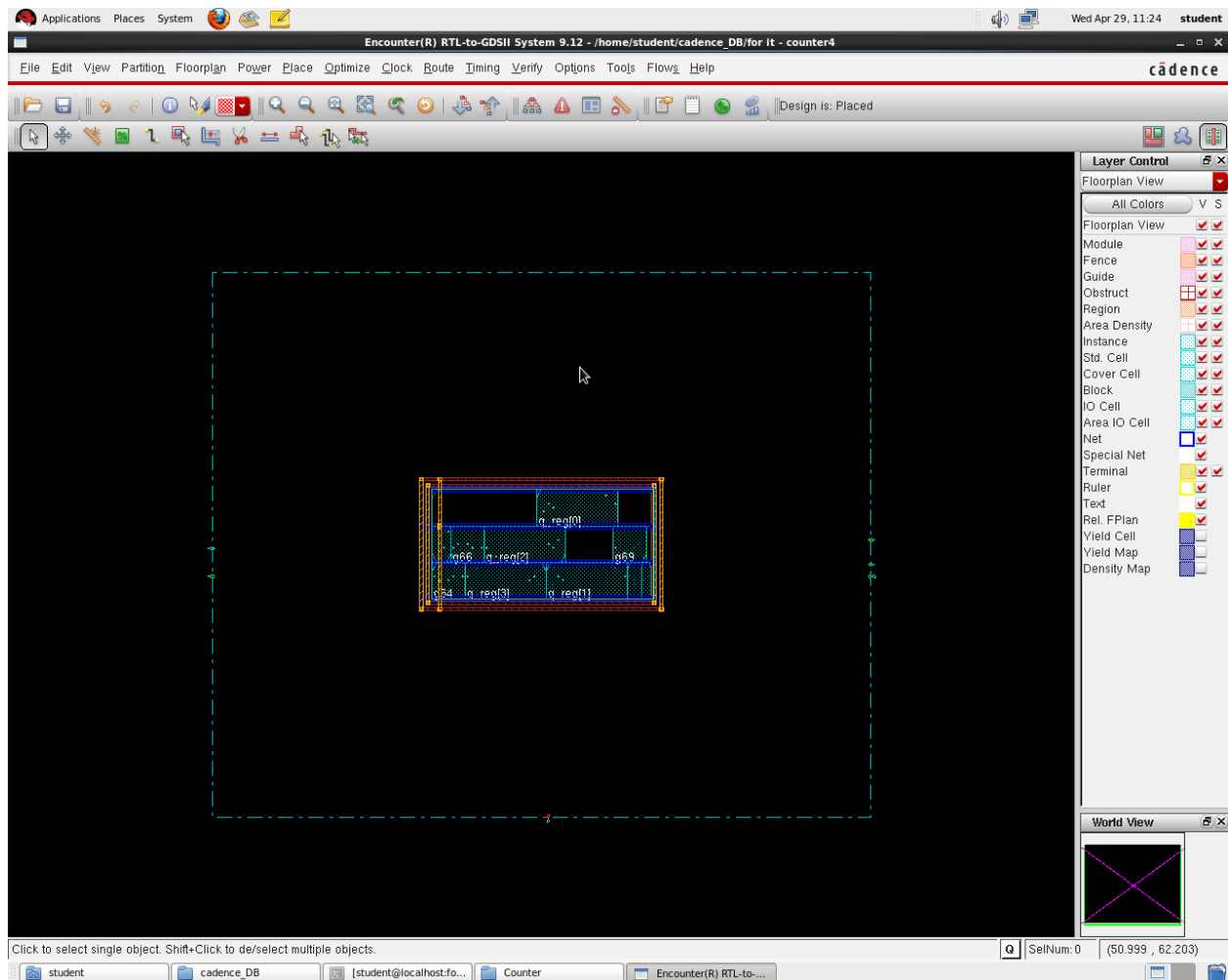
11

Go to Clock, click —Synthesize Clock Tree, a new window —Synthesize Clock Tree will get open. Click on Gen Spec and a new window —Generate Clock Spec will open.

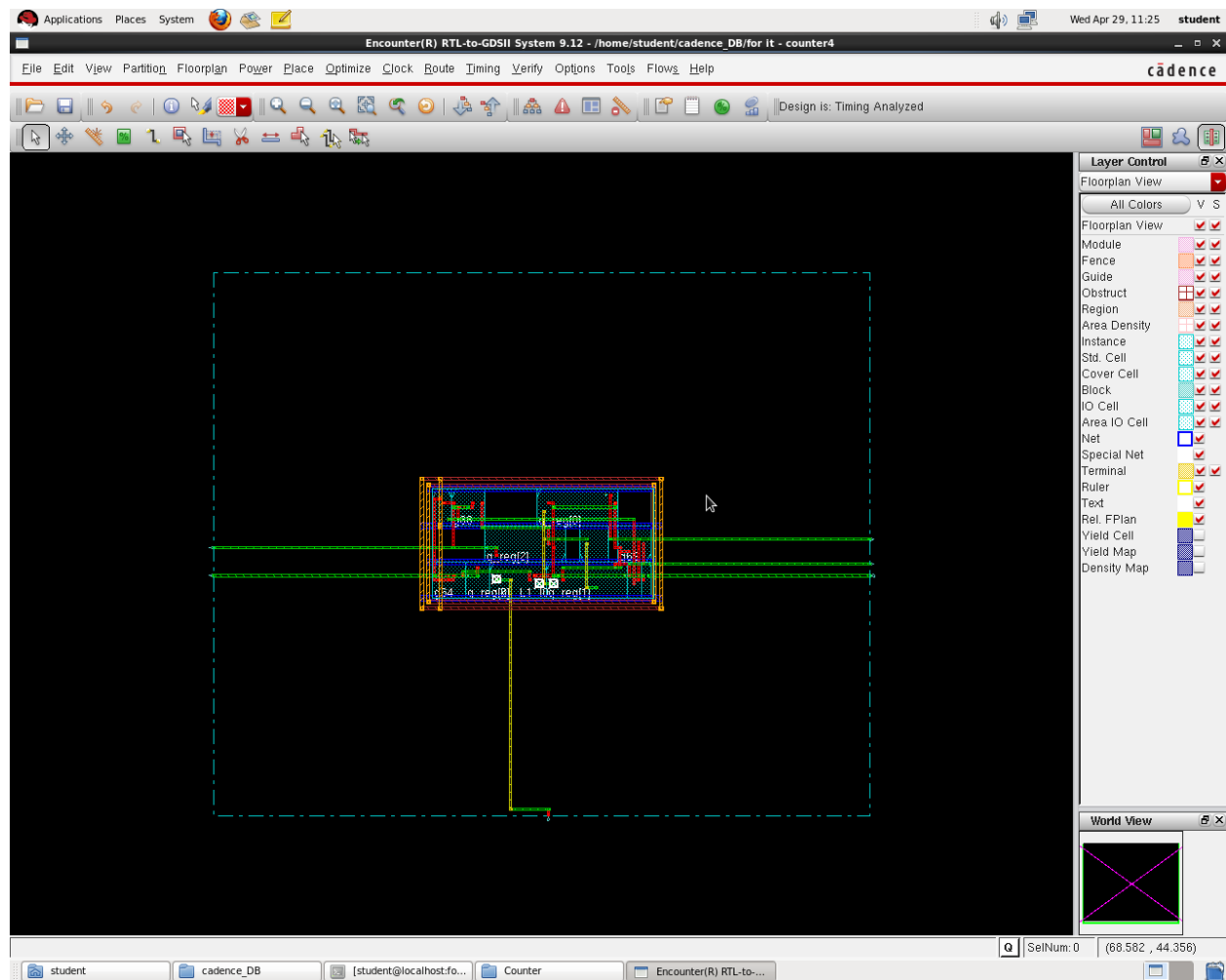
From Cells List, Select all clocks starting with —CLK and click on Add button to add them to the Selected Cells. Select a name for Output specification.

12

Again Perform the Timing by clicking on Timing and selecting Report Timing. Select —Post-CTS under Design Stage and do the select —Set-up as Analysis Type.



ENCOUNTER FINAL VIEW



INFERENCE

Using RTL compiler Netlist and sdc file is generated for the array Multiplier.

Date: 28.04.2015

ASIC Lab
Experiment No. 7

ASIC Implementation of Johnson Counter

Radheshyam Baviskar
14MVD1039

AIM: To design and realize Johnson's Counter for ASIC implementation

TOOLS USED: Cadence RTL Compiler 64-bit, Encounter

PROCEDURE:

1. Log into Unix Shell environment using **ssh** command via **terminal**
2. Source the required **sshrc** file from **/home/14mvd1039/cadence_DB** using **source sshrc** command
3. Navigate to the path **/home/14mvd1039/cadence_DB/Johnson** using command **cd /home/14mvd1039/cadence_DB/Johnson/johnsoncount**
4. Write the following timing constraint and save it as file **counter_constraint.g**:

```
create_clock -name clk_166mhz -period 6 -waveform {0 3} [get_ports "clk"]
set_clock_transition -rise 0.1 [get_clocks "clk_166mhz"]
set_clock_transition -fall 0.1 [get_clocks "clk_166mhz"]
set_clock_uncertainty 0.1 [get_ports "clk"]

set_input_delay -max 1.0 [get_ports "reset"] -clock [get_clocks "clk_166mhz"]
##set_input_delay -min 0.1 [get_ports "reset"] -clock [get_clocks "clk"]

set_output_delay -max 1.0 [get_ports "q"]
```
5. Invoke RTL Compiler tool using **rc -64** command inside the directory structure which holds the RTL code that needs to be synthesized (here **/home/14mvd1039/cadence_DBs/Johnson**)
6. The following commands are to be used for synthesis:
 - a. **set_attribute source_verbosity true**
 - b. **set_attribute information_level 9** – switch on all the **information** levels
 - c. **set_attribute library library/slow.lib** – load the required **library** file (slow.lib or fast.lib)
 - d. **read_hdl -v2001 bit4_counter.v** – read the RTL file for **syntax errors**
 - e. **set_top_design bit4_counter** – set the module in the RTL file as **top module**
 - f. **elaborate** – this creates a technology-independent schematic for bit4_counter
 - g. **read_sdc bit4_counter_constraint.g** – read the **clock constraint** file
 - h. **synthesize -to_mapped -effort low** – **synthesize** the RTL code with the effort setting as **low** (can either be **low** or **medium** or **high**)
 - i. **report power >bit4_counter_power.rep** – save the power report to the file **bit4_Johnson_power.rep**
 - j. **report area >bit4_counter_area.rep** – save the area report to the file **bit4_johnson_area.rep**
 - k. **report timing >bit4_counter_timing.rep** – save the timing report to the file **bit4_johnson_timing.rep**
 - l. **report qor >bit4_counter_qor.rep** – save the qor report to the file **bit4_johnson_qor.rep**

- m. gui_show – show the RTL schematic of the circuit using **Encounter** tool
7. From the timing report, make sure that timing slack is not negative. If it is, modify the timing constraint file (clock signal), follow the entire set of steps mentioned above and note down the timing slack again. Follow this procedure, till the timing slack obtained is positive

VERILOG CODE:

//----- 4BIT JOHNSON COUNTER---=-----//

```
module johnson(clk,rst,q);
input clk,rst;
output reg [3:0] q;

always @(posedge clk)
begin
if(rst==1'b1)
q<=4'b0;
else
begin
q[3]<= ~q[0];
q[2]<= q[3];
q[1]<= q[2];
q[0]<= q[1];

end
end
endmodule
```

4BIT COUNTER TEST BENCH---

```
module john_test;
reg clk,rst;
wire [3:0]q;

johnson uut(clk,rst,q);

initial
begin
clk=0;
rst=1;

#50
rst=0;

#200
$finish;
```

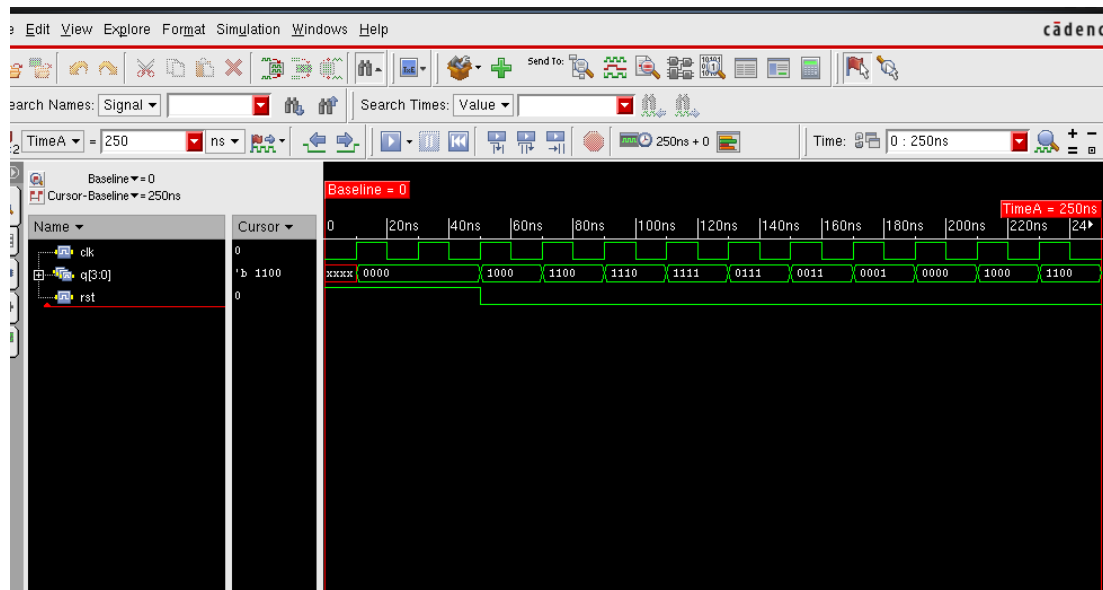
```
end
```

```
always
```

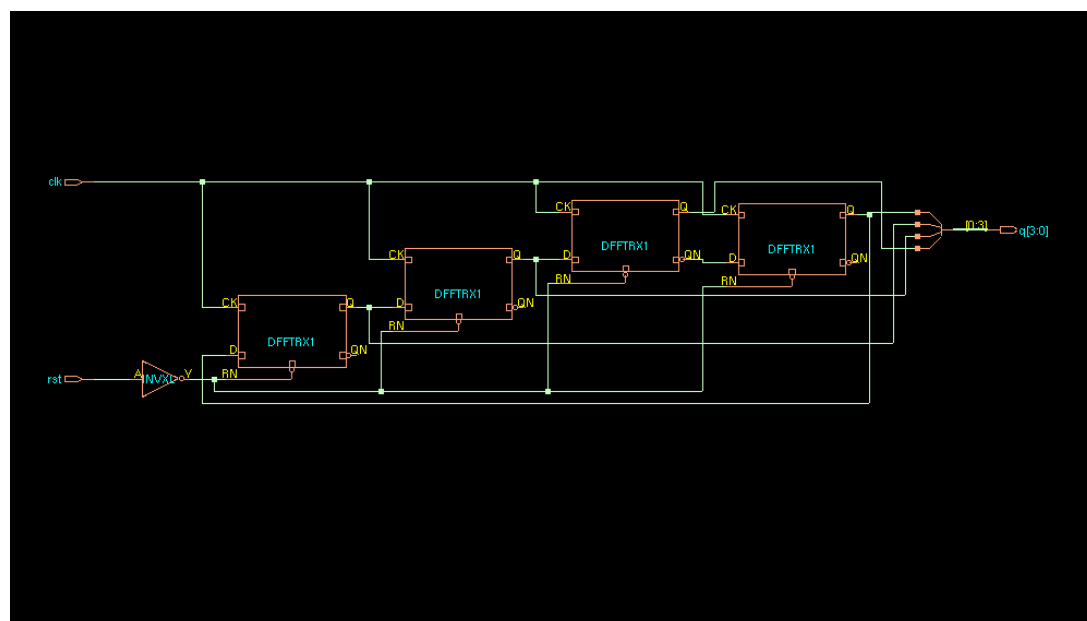
```
  #10 clk = ~clk;
```

```
endmodule
```

SIMULATION:



RTL SCHEMATIC:



REPORTS:

Area

```
=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Apr 29 2015  12:35:59 PM
Module:           johnson
Technology library: tsmc18 1.0
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====
```

Gate	Instances	Area	Library
DFFTRX1	4	226.195	tsmc18
INVXL	1	6.653	tsmc18
total	5	232.848	

Type	Instances	Area	Area %
sequential	4	226.195	97.1
inverter	1	6.653	2.9
total	5	232.848	100.0

Power

```
=====
Generated by:      Encounter(R) RTL Compiler v09.10-p104_1
Generated on:      Apr 29 2015  12:35:59 PM
Module:           johnson
Technology library: tsmc18 1.0
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====
```

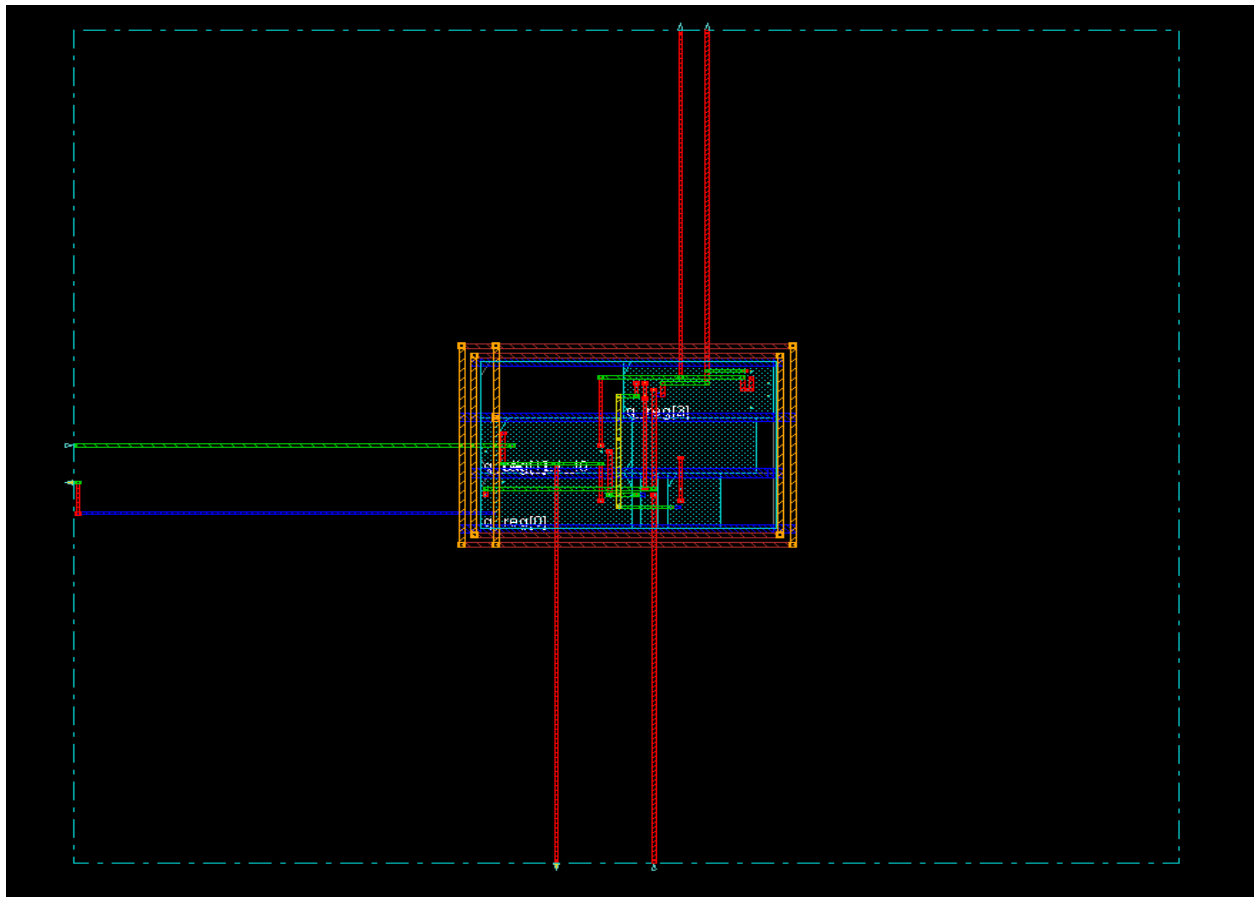
Instance	Cells	Leakage Power (nW)	Dynamic Power (nW)	Total Power (nW)
johnson	5	6.235	8061.465	8067.701

NETLIST:

```
// Generated by Cadence Encounter(R) RTL Compiler v09.10-p104_1

module johnson(clk, rst, q);
  input clk, rst;
  output [3:0] q;
  wire clk, rst;
  wire [3:0] q;
  wire n_1, n_7;
  DFFTRX1 \q_reg[0] (.CK (clk), .D (q[1]), .RN (n_1), .Q (q[0]), .QN
    (n_7));
  DFFTRX1 \q_reg[1] (.CK (clk), .D (q[2]), .RN (n_1), .Q (q[1]), .QN
    ());
  DFFTRX1 \q_reg[2] (.CK (clk), .D (q[3]), .RN (n_1), .Q (q[2]), .QN
    ());
  DFFTRX1 \q_reg[3] (.CK (clk), .D (n_7), .RN (n_1), .Q (q[3]), .QN
    ());
  INVXL g10(.A (rst), .Y (n_1));
endmodule
```

ENCOUNTER FINAL VIEW:



INFERENCE:

ASIC Design of Johnson Counter is done using Cadence tool, including nclaunch, RTL Compiler and Encounter tool.